ANNUAL REVIEWS

Annual Review of Astronomy and Astrophysics

Blue Stragglers and Friends: Initial Evolutionary Pathways in Close Low-Mass Binaries

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Annu. Rev. Astron. Astrophys. 2025. 63:467–512

The *Annual Review of Astronomy and Astrophysics* is online at astro.annualreviews.org

https://doi.org/10.1146/annurev-astro-071221-054402

Keywords

binary-star evolution, blue straggler stars, yellow straggler stars, sdB stars, mass transfer, mergers

Abstract

The scope of this review is the first stage in the evolution of close binary stars having components with $M < 2\ M_\odot$. An observational taxonomy for the products of such binary evolution is provided in the framework of dwarfs (blue straggler stars), giants (yellow straggler stars), subdwarf B stars, and giant-like stars (sub-subgiant stars and red straggler stars).

- Blue stragglers and yellow stragglers have directly measured masses greater than the main sequence turnoff masses of coeval populations. Observational evidence points to mass transfer as the most frequent formation channel for first-stage binary evolution products, occurring with enhanced stability and a range of mass-transfer efficiencies.
- Rapid rotation is an observed hallmark of products and an expected outcome of all proposed formation channels—mass transfer, mergers, and collisions. Excess angular momentum must be removed to permit observed mass gains by processes yet to be understood.
- Key theoretical issues remain. The stability of mass transfer from red giant and asymptotic giant branch donor stars remains ill-understood. Models struggle to account for the observed distributions of orbital eccentricities and periods. The loss of mass and angular momentum from a binary system is largely unconstrained. Detailed physical models for mergers of low-mass main sequence binaries are lacking.

First-stage binary evolution products constitute a substantial fraction of all evolved stars in old stellar populations. They travel along major alternative pathways of stellar evolution and in regions of the Hertzsprung–Russell diagram not populated by single stars.

Contents

1. INTRODUCTION

Our understanding of the evolution of single stars is one of the great intellectual accomplishments of the past century. This accomplishment relied heavily on star clusters: for the delineation of sequences in the properties of stars, as ongoing testbeds for ever more sophisticated models, and as exemplars for education and broader impact of the discoveries.[1] Ultimately, this understanding of single-star evolution set the foundation for understanding stellar populations and their role in evolutionary processes throughout the Milky Way galaxy and the Universe beyond.

Nonetheless, from these beginnings—e.g., the Sandage (1953) color–magnitude diagram (CMD) of the globular cluster M3 and the Johnson & Sandage (1955) CMD of the open cluster M67—cluster CMDs have shown many stars that do not lie along the single-star main sequence (MS) or giant branch and white dwarf (WD) sequences. The most famous of these are the blue straggler stars (BSSs), but there are numerous other instances as well. As research focused on the evolution of single stars, consideration of stars off these sequences as being anomalies or exotica became the norm, and use of this language continues to this day.

[1] Today, it is a rare textbook that does not show some version of figure 6 from Johnson & Sandage (1955).

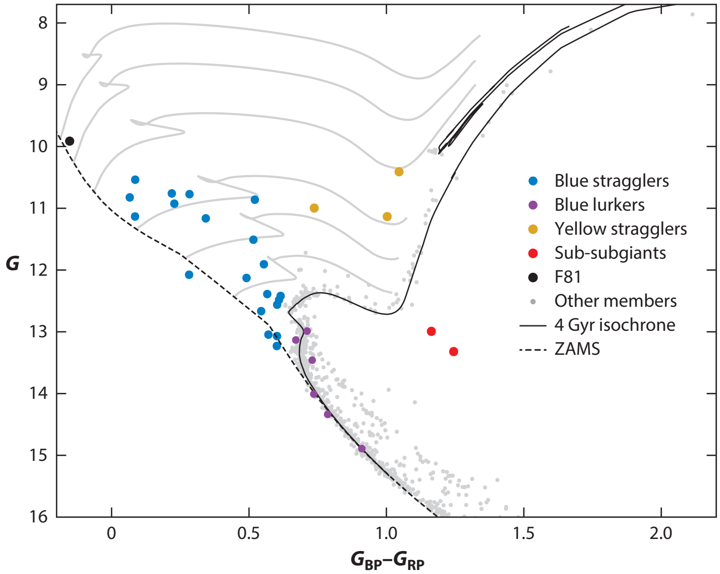

Figure 1

Color–magnitude diagram of members of the old open cluster M67, highlighting products of first-stage binary evolution discussed in this review. The fit isochrone is from Reyes et al. (2024). The MIST evolutionary tracks (Choi et al. 2016, *gray lines*) have masses, from bottom to top, of 1.5, 1.75, 2.0, 2.5, 3.0, and 3.5 $M_{\odot}$. Figure provided by Evan Linck. Abbreviations: MIST, MESA (Modules for Experiments in Stellar Astrophysics) Isochrones and Stellar Tracks; ZAMS, zero-age main sequence.

In fact, stars on these alternative evolutionary pathways are as much a norm as those on single-star pathways and, indeed, quite frequent. For example, in **Figure 1**, we show a modern CMD of the old (4 Gyr) open cluster M67, including only three-dimensional kinematic cluster members. More than 25% of the evolved[2] stars do not fall on the subgiant or giant branches. These stars are on frequently traveled alternative pathways of stellar evolution.

At the same time, photometric and especially spectroscopic studies of Galactic field dwarfs and giants identified a wide array of stars with properties not easily explained by single-star evolution. Examples included barium (Ba) stars, carbon-enhanced metal-poor stars, dwarf carbon (dC) stars, subdwarf B (sdB) stars, and field BSSs. The variety of types of such stars identified became so extensive that keeping track of them became an accomplishment in itself.

Crucially, with time-series radial-velocity and photometric observations, most of these stars—in both star clusters and the field—have been identified to be in binary stars with orbital periods of less than tens of thousands of days (e.g., McClure 1983, Carney et al. 2001, Lucatello et al. 2005, Mathieu & Geller 2009, Giesers et al. 2019). Indeed, it has become clear that all of these stars are on the evolutionary pathways of the many solar-like stars formed in binaries with periods of less than $\sim 10^5$ days (e.g., Raghavan et al. 2010).

In parallel with this observational journey, extensive work was being done by theorists to understand the inevitable phenomenon of mass transfer in binary stars, driven by the evolution and expansion of the primary star (e.g., Paczyński 1971, Webbink 1985). In part, this work was motivated by the need to explain an equally wide array of binary stars in advanced evolutionary states that ran counter to single-star evolution, including Algols, symbiotic stars, cataclysmic variables,

[2]Here, "evolved" is intended broadly to mean stars that photometrically are not on the single- or binary-star MSs.

novae, and double WD binaries. Soon, theorists also brought stellar mergers and collisions into the discussion (McCrea 1964, Nelson & Eggleton 2001, Sills et al. 2005).

Today, binary stars are recognized as the progenitors of stars found across nearly the entire Hertzsprung–Russell diagram (HRD) and are cited as the origins of a much wider extent of phenomena such as Type Ia supernovae and gravitational wave sources. The overarching goal must be to find integrated understandings of both observation and theory within which all of these binary evolution products can be understood. This goal is a grand challenge for binary star research, as significant goals should be.

We are not there yet, and today no single review can encompass all of the binary evolutionary pathways in detail. The scope of this review is the evolution of close MS binary stars having components with $M < 2\,M_{\odot}$ and, specifically, the first stage of their evolution that often—but not always—is driven by the internal evolution of the primary star. We define this first stage of evolution as continuing through the giant phases of the companion star, prior to a possible second mass-transfer episode. The recent review of the evolution of massive-star binaries by Marchant & Bodensteiner (2024) is a natural complement to this review.

2. A TAXONOMY OF LATE-TYPE BINARY EVOLUTION PRODUCTS

2.1. Conceptual Framework

The products of alternative evolutionary pathways typically have been stars identified by a variety of differences from single stars, in CMD location, surface abundances, rotation, hot companions, asteroseismology, etc. As such, they have not been brought together within a single holistic classification scheme, and their mutual associations have often been missed. We seek to provide such a holistic scheme in order to actualize their physical associations.

Most of these products have been found to be in binary stars themselves, often after their initial identification. Although the progenitor binary evolution is frequently initiated by evolution of its primary star, the identified product often is the secondary star.

Our conceptual foundation is that both stars in these binaries have gained or lost mass during their MS or giant evolution due to interaction with their companions. Our organizing framework encompassing all of these first-stage products is mass gainers and mass donors. Mechanisms for such mass changes are discussed below, but broadly the reader might keep in mind Roche-lobe overflow (RLOF), wind mass transfer, common envelopes (CEs), stellar mergers, and stellar collisions. Finally, these mass changes also reset the component ages. See the sidebar titled The Many Ages of Binary Evolution Products.

Finally, we restrict the progenitor binary components to MS stars with $M < 2\,M_{\odot}$. We note that such MS stars have giant evolution leading to degenerate He cores, and most have convective envelopes from birth.

THE MANY AGES OF BINARY EVOLUTION PRODUCTS

Mass gainers are sometimes described as “rejuvenated” stars. The addition of mass leads to ambiguity in describing the ages of binary evolution products. For future clarity, we propose the following:

- **Formation age:** Time since the formation of the progenitor binary.
- **Transformation age:** Time since completion of formation of the binary evolution product.
- **Evolution age:** Age associated with the interior structure, determined by comparison with the evolution of a single star of the product mass.

These concepts are adapted from Sun et al. 2021.

2.2. Observational Taxonomy

Although this conceptual foundation is powerful for developing organization and physical understanding, theory in the abstract cannot yet predict all of the products of such mass changes. An observational taxonomy is needed for stars that cannot be explained by single-star evolution theory. We recommend a taxonomy founded on location in the HRD, supplemented as needed with other stellar properties. Rather than inventing a new lexicon, we condense the current terminology in the literature into domains in the HRD, with our continued use of "straggler" as an intentional nod to the early explorers in this field. We start each definition with the current understanding of the physical state of products in each domain, but the definitions themselves are purely observational. The reader may find **Figure 1** useful while reading the following.

Blue straggler star (BSS): a star that has gained mass during its main sequence evolution and currently continues to burn hydrogen in its core

Yellow straggler star (YSS): a star that has gained mass during its main sequence evolution and currently is undergoing post–main sequence evolution

Subdwarf B (sdB) star: a star that is core helium burning with only an extremely thin hydrogen envelope, yielding observable temperatures of tens-of-thousand degrees

Sub-subgiant star (SSG): a star that is fainter than a subgiant and redder than main sequence single and binary stars

Red straggler star (RSS): a star to the red of the red giant branch and brighter than the base of the red giant branch

2.2.1. Blue straggler star.

We define BSSs broadly as stars that have gained mass during their MS evolution and currently continue to burn hydrogen in their cores.

Historically, BSSs were typically identified as stars brighter and bluer than the main sequence turnoff (MSTO) of an associated coeval population. This was an overly exclusive definition, as there are stars that are bluer but not brighter than the associated MSTO (even when the turnoff is defined by overshoot evolutionary models; VandenBerg & Stetson 2004). We include as "classical" BSSs those stars between the zero-age main sequence (ZAMS) and terminal-age main sequence (TAMS) whose locations distinguish them from single stars or photometric binaries in the associated coeval population. We note that this is a physical definition; previous studies have often defined the red boundary of BSSs by a color limit in the optical (if it was defined at all).

However, this definition is still too limited to include all of the mass gainers within the broader definition of BSSs above. For example, MS stars can gain mass without their final luminosities rising above the MSTO. Such stars can be embedded within the MS or among photometric binaries above the MSTO. Candidates for such stars have been identified by Leiner et al. (2019), who called these stars blue lurkers (BLs). Furthermore, in the Galactic field a well-defined coeval MSTO population with a single metallicity is seldom available. Thus, a complete census of BSSs requires secondary observable indicators of mass changes. For the discovery of BLs, Leiner et al. (2019) used rapid rotation in the absence of tidal locking by a nearby companion. In the field, indicators are often surface abundances, evidence of accretion from an evolved companion, or the presence of a hot WD companion.

We have not placed an upper brightness limit in this definition. In some clusters, BSSs have been identified more then two magnitudes above the MSTOs, perhaps most famously F81 in M67 (**Figure 1**), although the *Gaia* parallax measurement makes its membership uncertain. Such luminosities correspond to single-star masses at or above twice the associated MSTO mass, presumably requiring interactions of more than two stars. When they are secure cluster members, such BSSs are significant challenges for theory.

Finally, we recognize the reviews of classical BSSs by Stryker (1993), Bailyn (1995), and Wang & Ryu (2024), and a rich set of topical reviews in Boffin et al. (2015).

2.2.2. Yellow straggler star.

We define YSSs as stars that have gained mass during their MS evolution and currently are undergoing post-MS evolution.

In the HRD, YSSs are stars that lie to the red of the TAMS and above the subgiant branch. This definition is intentionally inclusive of products embedded within the red giant branch (RGB), horizontal branch (HB) or red clump (RC), and asymptotic giant branch (AGB) of an associated coeval population. However, as with the BLs, YSSs embedded in these evolved sequences require secondary indicators of prior mass gain. Historically, surface abundances have played this role, with the Ba giants being exemplars. More recently in open clusters, asteroseismology has revealed stars on evolved sequences with masses far above the MSTO mass (Section 3.3).

It is worth noting that stars in the YSS domain of a CMD can also be produced by the composite light of two stars found on other sequences in the cluster (e.g., a MSTO star and a giant), especially in the Hertzsprung gap. In recent literature, these too have been called yellow stragglers (e.g., Sales Silva et al. 2014). Inclusion of such interlopers in the YSS classification is not our intention here. Care must be taken to exclude such cases when studying individual objects. The most secure YSSs do not permit such origins of their photometry, unless the YSSs are confirmed spectroscopically.

2.2.3. Subdwarf B star. sdB stars are core He burning with only an extremely thin hydrogen envelope, yielding their defining observed high temperatures.

These stars have effective temperatures of tens of thousands of degrees but HRD locations below the ZAMS. (Subdwarf A stars have somewhat cooler effective temperatures.) Although historical discussions convolved them with extreme HB stars resulting from extreme mass loss at the RGB tip, more recent evidence points to most sdBs being binary primary stars that have lost most of their envelopes via mass transfer during late-RGB evolution. As such they are mass donors.

Recent reviews (Heber 2009, 2016; Lynas-Gray 2021) preclude the need for extensive discussion of the stellar characteristics of sdBs here. Thus, we limit discussions to their orbital properties and population characteristics in star clusters and the field.

2.2.4. Red straggler star and sub-subgiant star. Over the years, the term red straggler star (RSS) has been used for stars in a wide variety of locations in the post-MS domain of the HRD. Geller et al. (2017a) attempted to clarify the nomenclature by limiting the use of "red straggler" to stars that are to the red of the RGB and brighter than the base of the RGB. This definition is well defined for given passbands, although a star can move in and out of the RSS domain for different choices of passbands.

In recent literature, there has been growing ambiguity in references to stars as RSSs or sub-subgiant stars (SSGs). The first use of "sub-subgiant" in the literature was by Belloni et al. (1998) for two stars in the open cluster M67 located below the subgiant branch (see **Figure 1**), and the first detailed discussion of these prototypes was provided by Mathieu et al. (2003). However, Albrow et al. (2001) called stars in the same CMD domain of the globular cluster 47 Tuc "red stragglers." Again, Geller et al. (2017a) attempted to clarify the nomenclature by distinguishing SSGs in a given population as stars that are fainter than subgiants and redder than MS stars (including MS photometric binaries). More recently, however, Leiner et al. (2022) have discussed a large population of field stars that they call SSGs, even though many lie to the red of the RGB of a 14-Gyr isochrone.

As these stars draw more and more attention, this nomenclature confusion needs resolution, but there is not yet enough clarity to do so here. Anticipating an evolutionary connection, we associate both populations in this taxonomy, distinguishing them by the HRD locations suggested by Geller et al. (2017a). We look forward to a clarified nomenclature in the near future as the origins and possible relationships of these two subgroups become better understood.

Finally, we note that this definition of RSSs and SSGs makes no presumption about whether mass has been previously gained or lost by these stars, which is largely due to the lack of fully developed theories for their origins (Section 4.4). We may yet find that formally many do not fit in the conceptual framework of this review.

2.2.5. White dwarf. In the context of this review, WDs typically arise as the remnant cores of evolved primary stars. Because mass transfer may arise at essentially any time during the post-MS evolution of a primary, the masses of WD companions may range from extremely low-mass (ELM) He WDs to fully developed C/O WDs.

Until recently, most binary evolution products have not been first identified as the WD component. WD companions were found subsequently, often via blue optical spectra or ultraviolet (UV) flux detections. However, the product definitions above, though historically associated with optical data, are not intended to be restricted to such. And indeed, recent wide-field or all-sky UV surveys have begun to identify products first through UV excesses. Notable examples of this are *Ultraviolet Imaging Telescope* (UVIT) imaging (Section 3.4.3) and the mining of GALEX (*Galaxy Evolution Explorer*) data in the field (Rebassa-Mansergas et al. 2017, Anguiano et al. 2022).

3. OBSERVED PROPERTIES OF FIRST-STAGE BINARY EVOLUTION PRODUCTS

The scope of this review is the first stage in the evolution of close binary stars in which both companions are MS stars with $M < 2\,M_{\odot}$. In the context of stable mass transfer, this first stage of binary evolution is initiated by the evolution and expansion of the primary star (the mass donor) and the subsequent transfer of mass to the secondary star (the mass gainer). During this process, mass may also escape from the binary system. For longer-period progenitor binaries, this mass transfer continues until only the core of the initial primary star remains. For shorter-period progenitor binaries, orbital evolution (e.g., due to magnetic braking or dynamical effects of tertiary stars) as well as expansion of the primary star can lead to a merger with a more massive star as the product, possibly also having a wide unevolved companion. Finally, collisions of stars can occur, in particular in binary resonant encounters in star clusters and in unstable multiple systems, again leading to a more massive star possibly having one or more companions.

A comprehensive review of the observational literature is provided in the **Supplemental Appendix**, organized by our taxonomy of binary evolution products within three distinct environments—open star clusters, globular star clusters, and the Galactic field. Here, we provide an integrated perspective on the common and contrasting astrophysical properties of these products, highlighted with specific observational results as exemplars.

Supplemental Material >

3.1. Population Frequencies

Binary evolution product frequencies are best determined via unbiased surveys, preferably in coeval populations. The *Gaia* data releases have yielded important advances in both completeness and quality of cluster proper-motion membership determination, especially in open clusters. Rain et al. (2021) identify 897 BSS candidates in 111 of 408 open clusters. Rain et al. find a rapid increase in the BSS specific frequency (relative to the number of MSTO stars) for clusters older than $\log t\,(\mathrm{year}) > 8.7$ (**Figure 2**), and indeed conclude that BSSs are absent in very young open clusters (however, see the review of Wang & Ryu 2024). Rain et al. also identified 77 YSS candidates in 43 open clusters, with most host clusters having only one YSS. Importantly, the frequency of BSSs in old clusters is substantial, a few percent to 20% of the number of stars within 1 mag of the turnoff (**Figure 2**). Even without accounting for the presumably shorter lifetimes of BSSs compared to MSTO stars, this census within open clusters shows that in the mass domain of this review BSSs are a frequent pathway of stellar evolution.

BSSs and YSSs in the Galactic field identified from elevated Ba abundances (also known as Ba stars) comprise ≈1% of their parent populations (respectively, F-type MS stars and G- and K-type giants) (MacConnell et al. 1972, North & Duquennoy 1991). Although small, this frequency is consistent with expectations from binary population synthesis (BPS) modeling of their formation through AGB mass transfer (e.g., Han et al. 1995, Izzard et al. 2010). The frequency of C-rich YSSs (also known as CH stars or CEMP-*s* stars) among metal-poor giants is not well determined but certainly much higher (e.g., Arentsen et al. 2022). BPS models in this case also predict high

Barium (Ba) star: FGK-type star with strong barium absorption lines

CH star: star with strong CH molecular bands, but warmer than a classical carbon star

CEMP-*s* star: a very metal-poor star with large surface carbon and *s*-process enhancements

Dwarf carbon (dC) star: cool dwarf star with molecular bands of C_2, CH, and CN

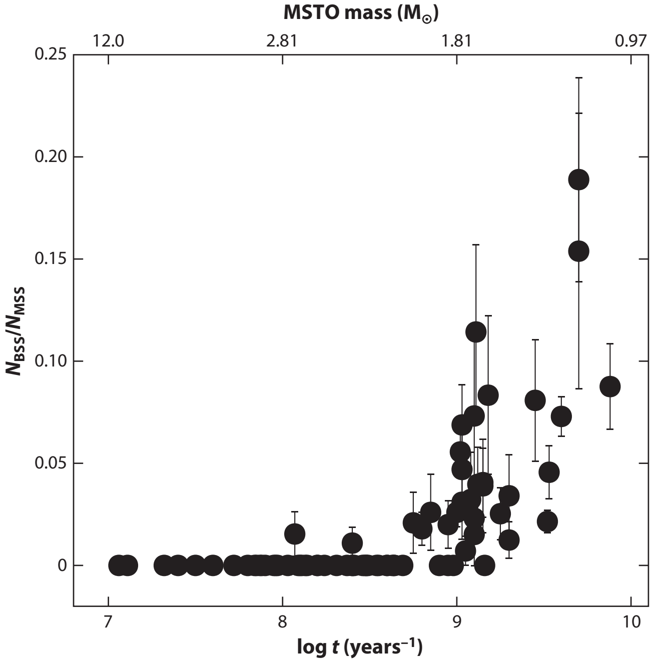

Figure 2

Number of BSSs as a function of open-cluster age (in units of log year), relative to the number of main sequence stars within one magnitude of the MSTO. Uncertainties are Poissonian. Also shown are the MSTO masses. Figure adapted from Rain et al. (2021); copyright ESO with permission from A&A. Abbreviations: BSS, blue straggler star; MSTO, main sequence turnoff.

frequencies up to about 10% (Abate et al. 2015b). In both instances, these abundance-selected products are tracers of larger field BSS and YSS populations.

Product frequencies are observed to vary between environments. For example, in the Galactic halo BSSs occur with a specific frequency relative to HB stars of ≈4, which is again evidence of frequent BSS formation (Preston & Sneden 2000). However, in globular clusters among stars with similar masses and metallicities, the BSS specific frequency relative to HB stars is lower by as much as an order of magnitude (Piotto et al. 2004). Frequencies also vary between globular clusters. Preston & Sneden (2000) find that BSS populations are systematically enhanced in lower-density globular clusters. (Observed environmental effects in globular clusters are further discussed in Section 3.7.)

As a final example of environmental variations, Geller et al. (2017a) find that the specific frequency of SSGs increases with lower-mass clusters. In particular, SSGs are far more frequent in open clusters than in globular clusters (as also are binary stars).

3.2. Color–Magnitude Diagrams

There are many CMDs in the literature showing the presence of BSSs and other binary evolution products. As exemplars, we draw from Leiner & Geller (2021), who integrate the CMDs of 16 old (1–10 Gyr), nearby (d < 3,500 pc) open clusters with masses greater than 200 $M_{\odot}$. The age range of these open clusters corresponds to MSTO masses of 1–2 $M_{\odot}$. **Figure 3** shows the cluster CMDs combined within three age groups. In addition to the evident classical BSSs, candidate YSSs and SSGs can be seen in all three age domains.

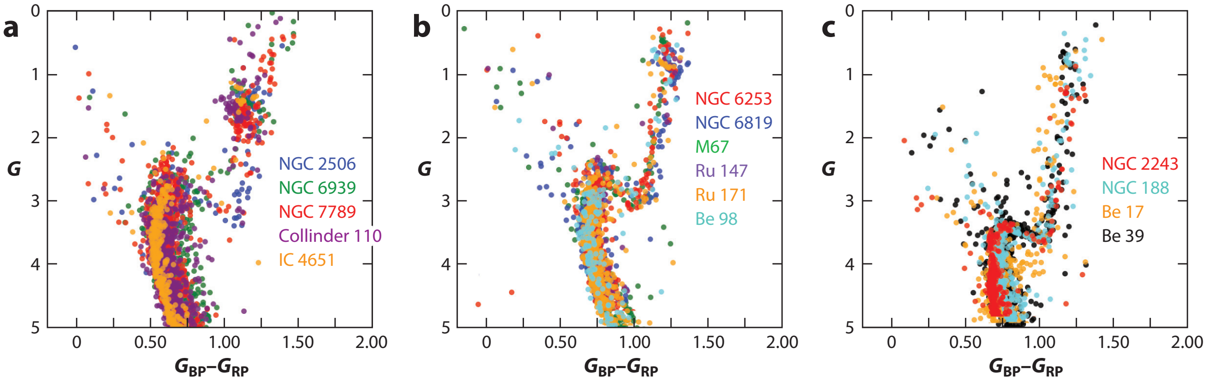

Figure 3

Combined color–magnitude diagrams of (*a*) young open clusters (1–2 Gyr), (*b*) intermediate-aged open clusters (2–4 Gyr), and (*c*) old open clusters (>4 Gyr). Each shows absolute *G* magnitude versus dereddened $G_{BP} - G_{RP}$ color. Small color and magnitude shifts have been applied to align the main sequence turnoffs of all clusters in each plot. Figure adapted from Leiner & Geller (2021) with permission from the AAS.

Leiner & Geller suggest that the classical BSSs in the older (>2 Gyr) clusters can be split empirically into two groups, divided by a gap in the BSS distribution at 0.5 $M_\odot$ above the MSTO mass. They find the distribution of BSS masses to be skewed toward lower masses (where masses are derived from MIST single-star evolution models). In addition, they find only 5 BSSs with masses more than 1 $M_\odot$ above their associated cluster MSTO masses, and none with masses more than 1.5 $M_\odot$ above the MSTO. The latter represents fully conservative mass transfer in the young clusters; thus, these authors note that none of the BSSs have masses greater than can be produced by binary mass transfer, which would otherwise require merger or collisional formation or multiple events. Using the `COSMIC` BPS code (Breivik et al. 2020) to model the mass-transfer formation channel, Leiner & Geller find that both enhanced RLOF stability and varying mass-transfer efficiency are needed to reproduce the global frequency and shape of the BSS mass distribution.

We emphasize that most classical BSSs do not fall on the ZAMS of their respective coeval populations, including BSSs with very small transformation ages (see also **Supplemental Figure 2**). In the context of single-star evolution theory, this reveals varying He abundances in their cores. As such, their CMD distribution bears information on the evolutionary states of their progenitor secondary stars in the mass-transfer formation scenario or of their progenitor primary stars in the merger or collision formation scenarios, as well as on relative formation rates.

Supplemental Material >

Classical BSSs lie in photometrically distinguishing locations of CMDs. However, lower-mass BSSs may be embedded photometrically within MSs as BLs and, thus, require additional diagnostics for identification. Rapid rotation is predicted by all mass transfer, merger, and collision formation mechanisms. Leiner et al. (2019) identified 11 BL candidates in M67 as rapidly rotating MS stars (see **Supplemental Appendix Section 1.2.1**). Six of these stars are BLs securely embedded within the M67 MS (**Figure 1**). These BLs have much in common with the BSSs, including a high fraction of long-period spectroscopic binary companions; circular to modest-eccentricity orbits; and, in one case, a hot WD companion (Nine et al. 2023, Leiner et al. 2025). Dattatrey et al. (2023a) identified four far-ultraviolet (FUV)-bright stars at the MSTO of the globular cluster NGC 362, which they interpret as BLs with WD companions. dC stars in the Galactic field are another example of likely mass gainers embedded in the lower MS, in this case identified as BLs by their C-rich surface chemistry.

In truth, distinguishing classical BSSs and BLs is likely a distinction without a fundamental difference. Perhaps the most significant takeaway from examination of **Figure 1** is the nearly continuous distribution of BSSs from the historical domain of classical BSSs to blueward of the overshoot blue hook at the MSTO and finally onto the unevolved MS. There may be a gap within the classical BSSs, as noted by Leiner & Geller (2021). There also is a reduced dispersion in color with lower luminosity on the MS, which is likely indicative of an increased uniformity of interior structures as would be expected of lower-mass progenitor mass gainers. At a more fundamental level, the entire BSS sequence is the convolution of distributions of initial masses, amounts of mass gained, and perhaps formation mechanisms.

YSSs, presumed to be evolved BSSs, are found in both cluster and field populations. Their distributions in CMDs have not been as extensively studied as for BSSs. Some are evident (e.g., in M67; see **Figure 1** and discussion in **Supplemental Appendix Section 1.3.1**). Others lie at the boundaries of binary sequences, RCs and HBs, making their photometric identification challenging (e.g., Linck et al. 2024).

Supplemental Material >

In globular clusters, the domains of YSSs and extended HBs overlap. Using *Hubble Space Telescope* (HST)/WFPC2 imaging, YSSs have been identified as being slightly bluer than the RGBs and between 0.2 and 1 mags brighter than the HBs in those clusters (Ferraro et al. 1999) (**Supplemental Figure 6**). The radial distributions of these candidates mimic those of the BSSs, at the least indicative of similar masses. Other YSS populations have been suggested through dynamical or population analyses (Beccari et al. 2006, Parada et al. 2016a, Kravtsov & Calderón 2021). Statistical estimates of YSS lifetimes based on population numbers are similar to single-star evolution lifetimes at the same masses.

As with BSSs, YSSs are sometimes embedded within single-star evolved populations and require additional diagnostics. Most of the Ba-selected YSSs in the field occupy the RC region of the HRD (**Supplemental Figure 8**), which is consistent with this being a relatively long-lived phase compared to that for the RGB. Over-massive stars are found within open cluster RGBs and RCs (Section 3.3). Based on their frequency in NGC 6819, Handberg et al. (2017) conclude that about 10% of all RGB stars have not evolved as single stars.

3.3. Masses

Very often in the literature the masses of binary evolution products are determined from single-star evolutionary tracks, on the presumption that, after a thermal relaxation time since their formation, these stars settle into interior structures like single stars with their masses and core compositions. Innumerable cluster CMDs in the literature have such tracks superimposed (e.g., **Figure 1**). In the field, Carney et al. (2005) used isochrone fitting to estimate the masses of halo BSSs, obtaining values between 0.83 $M_{\odot}$ (just above the halo turnoff mass) and 1.28 $M_{\odot}$. Escorza et al. (2017, 2019b) found similar masses of 0.8 $M_{\odot}$ to 1.5 $M_{\odot}$ for field Ba- and CH-selected BSSs (see **Supplemental Figure 8**). The isochrone mass distribution of Ba-selected YSSs begins at 1 $M_{\odot}$ and peaks around 2 $M_{\odot}$ (Escorza et al. 2017). In an interesting twist, the physics of stellar interior models have been modified for rapid rotation so as to fit SSGs and RSSs, and thereby determine their masses (e.g., Leiner et al. 2017, Stassun et al. 2023).

Unfortunately, few binary evolution products have highly precise, dynamically measured masses to test this application of single-star models. Precise mass measurements have been made asteroseismically for several YSSs on the giant branches of open clusters in the *Kepler* field (**Supplemental Appendix Section 1.3.2**). Handberg et al. (2017) found asteroseismic masses of 2.4–2.6 $M_{\odot}$ for two stars near the RC of NGC 6819, with relative precisions of 4–10%. These are to be compared with asteroseismic masses of $\approx$1.6 $M_{\odot}$ for other NGC 6819 RC stars and an isochronal MSTO mass of 1.4 $M_{\odot}$. Handberg et al. (2017) also find in NGC 6819 that a luminous single RGB (or AGB) star is even more overmassive at 2.80 $\pm$ 0.25 $M_{\odot}$, which they suggest is an evolved BSS of merger origin. Finally, Leiner et al. (2016) used asteroseismology to measure at similar precision a mass of 2.9 $M_\odot$ for the most luminous YSS in M67 (**Figure 1**), which is more than twice the cluster MSTO mass of 1.3 $M_\odot$.

Comparable or less precise measures have been made for individual BSSs in open clusters and globular clusters via eclipsing binaries (van den Berg et al. 2001, Sandquist et al. 2003), pulsations of SX Phoenicis variables (Nemec et al. 1995, Gilliland et al. 1998, Fiorentino et al. 2014), surface gravity measurements (Shara et al. 1997, De Marco et al. 2005), and ellipsoidal variability (Brogaard et al. 2018) (see **Supplemental Appendix Sections 1.2.3 and 2.2.4**). The mass measurements from these various techniques agree to within 20% with masses from single-star evolutionary tracks, which themselves have systematic differences of ≈10%. In all cases these assorted measurement techniques yield classical BSS masses in excess of associated MSTO masses.

Supplemental Material >

Statistical mass estimates for populations of BSSs have also been made in clusters based on mass segregation (Mathieu & Latham 1986, Parada et al. 2016b), also finding masses in excess of the MSTO masses. Similarly, candidate YSSs have been found in globular clusters with radial distributions that mimic the BSSs in the same clusters (see **Supplemental Appendix Section 2.3**). Note that these stellar dynamical approaches measure the masses of the entire binary systems containing the BSSs.

Although the core presumption that these binary evolution products have gained mass is secure, highly precise and accurate dynamical masses and asteroseismology are needed to test and explore their interior structures.

3.4. Binary Properties

The orbits and compositions of binary evolution products are crucial clues to their formation and subsequent evolutionary pathways. Here, we focus on observations of their frequencies, orbital parameters, and companions.

3.4.1. Binary frequencies. The binary frequencies of binary evolution products are much higher than MS binary frequencies at similar masses. Complete radial-velocity surveys of classical BSSs in the old open clusters M67 and NGC 188 yield frequencies of binaries with $P < 10^4$ days of 80 ± 20–25%, to be compared with MS binary frequencies in the same clusters of ≈25% (Latham 2007, Mathieu & Geller 2009). For halo BSSs, Carney et al. (2001) find a binary frequency of 60 ± 24% for $P < 1{,}600$ days and derive a frequency of 68 ± 10% based on orbital solutions (and possible solutions) from Preston & Sneden (2000). These binary frequencies—though high—are not 100%. It remains an open question whether non-velocity-variable BSSs are in fact single or rather are in binaries of very long periods. Certainly there is currently room within the above binary frequencies for single products of merger or collision origin.

In contrast, Ba-selected YSSs in the Galactic field are essentially all detected spectroscopic binaries (Jorissen et al. 1998, 2019). This is consistent with the interpretation of these stars as the products of AGB mass transfer. The detected binary frequency among Ba- and CH-selected BSSs in the field is ≈67% (Escorza et al. 2019a), but the data for this sample are much less complete than for the Ba-selected YSSs.

For dC stars, Whitehouse et al. (2018) reported 21 radial-velocity variables among 28 monitored stars, which they find consistent with a 100% binary frequency. Roulston et al. (2019) reached a similar conclusion based on a larger radial-velocity survey of 240 dC stars.

3.4.2. Orbital periods and eccentricities. Eccentricity–period diagrams are shown in **Figure 4** for an array of first-stage binary evolution products in open clusters and the field. At a high level, the most striking observation from **Figure 4** is the similarity of the eccentricity–period distributions across all binary evolution products.

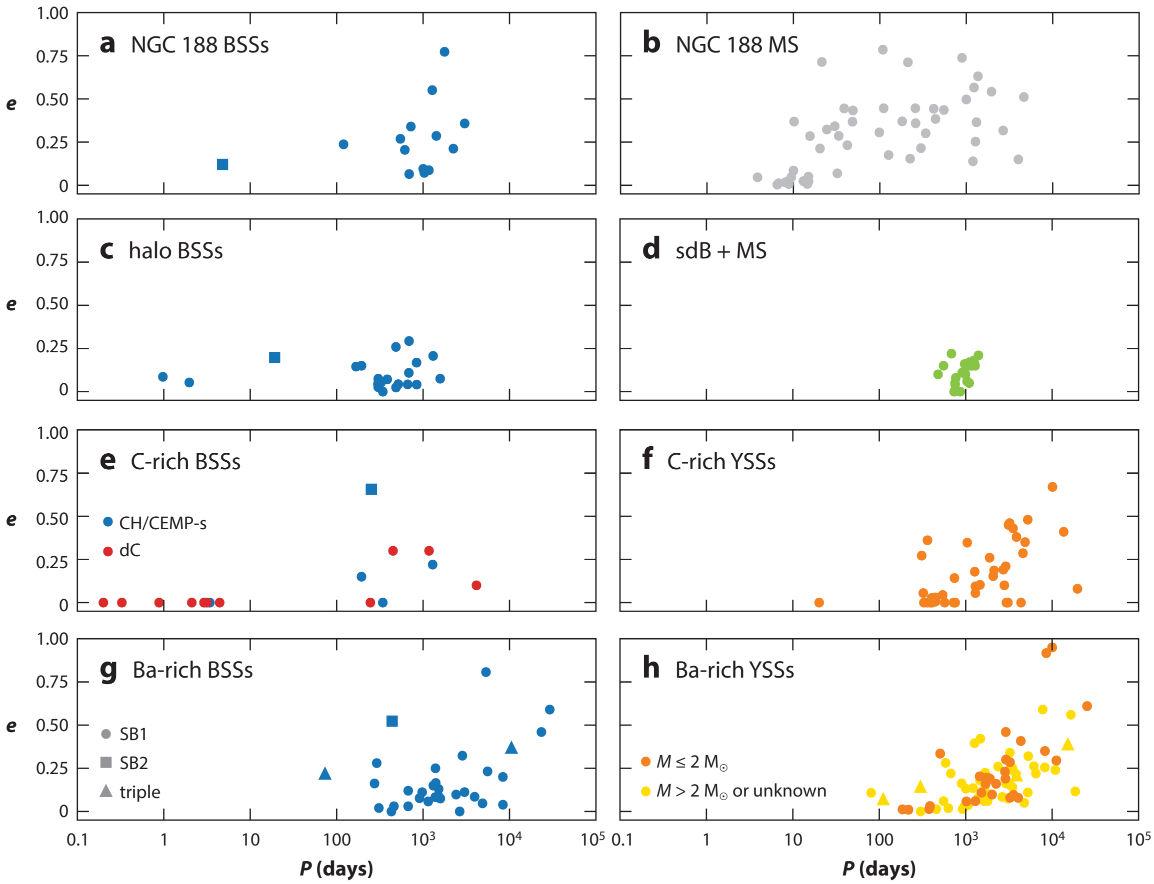

Figure 4

Periods and eccentricities for different samples of post-mass-transfer binaries: (*a*,*b*) the classical BSSs and MS binaries in NGC 188, (*c*) classical BSSs in the halo, (*d*) sdB + MS binaries, (*e*,*f*) C-rich BSSs and YSSs, and (*g*,*h*) barium-rich BSSs and YSSs. BSSs are color-coded in blue (except for the dC stars, which are of lower mass, in *red*), YSSs are shown in orange, and stars identified as mass donors are shown in green. Square symbols are double-lined spectroscopic binaries; triangles indicate the inner and outer orbits of triple systems. Data taken from (*a*,*b*) Mathieu & Geller (2015); (*c*) Carney et al. (2001, 2005), Sneden et al. (2003); (*d*) Vos et al. (2017, 2019); (*e*,*f*) Jorissen et al. (2016), Hansen et al. (2016), Harris et al. (2018), Roulston et al. (2021), Whitehouse et al. (2021); (*g*,*h*) Jorissen et al. (1998, 2019), Van der Swaelmen et al. (2017), Escorza et al. (2019b), North et al. (2020), and Escorza & De Rosa (2023). Abbreviations: BSS, blue straggler star; CEMP, carbon-enhanced metal poor; MS, main sequence; sdB, subdwarf B; YSS, yellow straggler star.

In the old (7 Gyr) open cluster NGC 188, Mathieu & Geller (2009) did a complete survey for binaries with $P < 10^4$ days among both the 21 BSSs and a large sample of solar-like MS stars. Comparison of panels *a* and *b* shows the BSS binary period distribution to be evidently different from that of the MS binaries. Most of the BSS binaries have orbital periods within a half-decade of 1,000 days, whereas the MS binaries in NGC 188 continuously populate all periods from a few days to 3,000 days (and to longer periods in other open clusters and the Galactic field). Another striking result is that the NGC 188 BSSs show a large dispersion of orbital eccentricities from $e = 0$ to $e \approx 0.8$ at long periods. On one hand, circular orbits are rare among long-period MS binaries, in both NGC 188 and other surveys (Raghavan et al. 2010). On the other hand, mass-transfer products historically have been presumed to have circular orbits.

Period distributions qualitatively similar to the NGC 188 BSSs are found for all of the binary evolution products in **Figure 4**, to within differences in long-period completeness. In more detail, most periods of the binary products range from about 100 to 40,000 days. For all products, a striking result in **Figure 4** is that the frequency of binary products drops precipitously at periods of ≈100 days. Longer orbital periods up to about 1,000 days can result from stable RLOF mass transfer from RGB donors (Section 4.2.3). However, many Ba- and C-rich BSSs and YSSs—which must result from AGB mass transfer—also have periods less than 1,000 days. Periods greater than ∼3,000 days exceed the maximum period for RLOF from AGB donors and are indicative of formation by wind mass transfer (Section 4.2).

All of the binary products—whether BSSs, YSSs, or sdBs—show dispersions of orbital eccentricities. Circular or modestly eccentric orbits dominate below ≈1,000 days, with a rising envelope of eccentricities at longer periods. The circular and near-circular BSS orbits may be indicative of dissipative processes during formation, and indeed circular orbits frequently are assumed to be the outcome in BPS models. But in fact most of the binary products have significant orbital eccentricities. This is an important observational challenge to binary evolution theory.

The eccentricities of the few very short-period orbits in **Figure 4** are also interesting. Several classical BSSs have small but significant eccentricities of unknown origin (as is also found in M67) (**Supplemental Appendix Section 1.2.2**). In marked contrast, all seven short-period dC binaries have circular orbits. This short-period dC population is absent among other products of AGB mass transfer and likely results from CE evolution (see Section 4.2.2.3).

Supplemental Material >

Subdwarf B binaries have two very distinct orbital period distributions. In the longest-period group, the orbital periods range between 500 and 1,400 days, and eccentricities are modest but again mostly nonzero (panel *d*). The companions are K- to F-type dwarfs. This limited period range of wide sdB binaries (and their rapid rotation) (Section 3.6) may be understood as resulting from stable RLOF initiated from donor stars near the tip of the RGB due to the small range of initial separations required to ignite He in an RGB star losing its envelope (Chen et al. 2013).

By contrast, sdB stars with low-mass M-dwarf (or brown dwarf) companions are found among a shorter-period population with periods of only 0.06 to 1.4 days (Schaffenroth et al. 2022). (Such short-period sdB binaries otherwise have WD companions, which result from more than one stage of mass transfer.) Like the short-period dC binaries, these short-period sdB stars are thought to form via a CE. Finally, in the Galactic field almost half of the sdB population is apparently single; mergers of He-core WDs have been suggested for the origin of such single sdBs (Heber 2016).

Post-RGB and post-AGB stars are in a short-lived evolution phase contracting toward the WD stage. Many of these stars are in binary systems with similar distributions of orbital periods and eccentricities to the binary evolution products discussed here (see **Supplemental Figure 9**). All are seen to have circumbinary disks (Van Winckel 2019). Their spectroscopic mass functions suggest low-mass MS companions, which implies that they are first-stage binary evolution products that may hold important clues to the mass-transfer phase from which they recently emerged.

3.4.3. Companions and companion masses. Companion detections provide important clues to formation scenarios. For example, mass transfer predicts remnants of the donor cores as companions to mass gainers. In fact, UV excesses and composite spectra indicative of WD companions are frequently found among BSSs and YSSs. In NGC 188, HST FUV photometry revealed hot WD companions to four long-period BSSs, with three more detected statistically (Gosnell et al. 2015). With effective temperatures higher than 13,000 K, the four BSSs with directly detected WDs have transformation ages of less than 250 Myr, which is extremely recent in comparison to the 7-Gyr age of the cluster. Two of the detections, including the BSS with the hottest WD, do not lie on the ZAMS (**Supplemental Figure 2**), implying that these BSSs formed recently with already evolved cores. Accounting for nondetections of cooler WDs, Gosnell et al. argue that on the order of two-thirds of the NGC 188 BSSs have WD companions.

Supplemental Material >

A great number of FUV detections have been made in 10 open clusters with UVIT, with UV excesses found among BSSs, YSSs, RC stars, and MS stars near MSTOs (BLs) (see **Supplemental Appendix Section 1.2.2**). In M67, UV excesses were found for 6 BSSs, 2 YSSs, and 3 MS stars (Pandey et al. 2021, and references therein). Again, many of the M67 BSSs with detected WDs lie off the ZAMS (**Supplemental Figure 3**). Similarly, Dattatrey et al. (2023b) used UVIT and archival FUV data to identify 12 FUV-bright BSSs in the globular cluster NGC 362. Panthi et al. (2023) observed 27 halo BSSs finding evidence for hot companions for 12 BSSs, 8 of which are known spectroscopic binaries. About a dozen dC stars show composite DA-type WD spectra in the optical (see, e.g., Green 2013).

Specific evidence for RGB mass transfer includes detection of a low-mass ($<0.5\ M_\odot$) He WD remnant. Gosnell et al. (2019) used FUV high-resolution spectra at Lyα to determine the masses of two WD companions to BSSs in NGC 188, with one having a mass of $0.42\ M_\odot \pm 0.02\ M_\odot$. Landsman et al. (1998) similarly measured a mass of $0.22\ M_\odot$ for the WD companion of a YSS in the open cluster M67. Both masses are indicative of He WDs. In the open cluster M67 and the globular cluster NGC 362, UV photometric mass determinations from WD evolution models yielded low-mass ($<0.3\ M_\odot$) He or ELM WD companions (Pandey et al. 2021, Dattatrey et al. 2023b). Subdwarf B stars are also examples of RGB mass transfer, albeit identified from the mass donor star rather than the mass gainer.

More massive C/O WD companions are indicative of AGB mass transfer. For the second NGC 188 BSS, Gosnell et al. (2019) measured a mass of $0.53\ M_\odot \pm 0.03\ M_\odot$ indicative of a C/O WD companion from an AGB donor. In the open cluster NGC 7142, Panthi et al. (2024) found several BSS companions with C/O WD masses determined photometrically. Escorza & De Rosa (2023) combined radial-velocity and astrometric data to measure companion masses for 60 field Ba-selected BSSs and YSSs and found most to be between $0.5\ M_\odot$ and $0.8\ M_\odot$. Roulston et al. (2021) measured a very hot (31,000 K), 0.57-$M_\odot$ WD from the Sloan optical spectrum of a dC-DA binary.

WD companion masses can be inferred statistically from orbital mass-function distributions. The complete sample of NGC 188 single-lined BSSs revealed a narrowly peaked mass distribution from $0.2\ M_\odot$ and $0.8\ M_\odot$ (Geller & Mathieu 2011), foreshadowing the He and C/O WD companions found spectroscopically. Carney et al. (2001, 2005) also find halo BSS secondary masses to be consistent with WD companions having a narrow mass distribution. Jorissen et al. (2016, 2019) found companion mass distributions to peak around $0.6\ M_\odot$ for field CEMP-*s*- and CH-selected YSSs and between 0.6 and $0.9\ M_\odot$ for Ba-selected YSSs, which is indicative of the C/O WD companions expected from AGB mass-transfer origins.

Importantly, MS companions have also been found among BSSs as double-lined spectroscopic binaries. Although infrequent (e.g., Preston & Sneden 2000, Carney et al. 2001 for halo BSSs), several are shown in **Figure 4**. The presence of an MS companion suggests formation by either a merger or collisional origin, possibly within a system with higher-order multiplicity (Section 4.3).

3.5. Surface Abundances

In the context of mass gainers, both field BSSs and YSSs are often identified by abundance excesses that trace material from AGB mass donors (see the sidebar titled Surface Abundances of Asymptotic Giant Branch Mass-Transfer Products). The observed heavy-element abundance patterns, when combined with detailed AGB nucleosynthesis models, can provide constraints on the masses of the AGB donors and on the amount of mass accreted by their companions.

For example, in 182 Ba-selected YSSs, de Castro et al. (2016) find a variety of *s*-process abundance patterns and a wide range of overall enhancements between [*s*/Fe] = 0.25 and 2.0. Stancliffe (2021) and Cseh et al. (2022) find that these abundances are best matched by AGB donors in the

SURFACE ABUNDANCES OF ASYMPTOTIC GIANT BRANCH MASS-TRANSFER PRODUCTS

The identification of BSSs and YSSs in the Galactic field can be hampered by the lack of a reference population of the same age and metallicity, so that identifying alternative evolution products from their locations in a CMD is not possible. However, mass gainers in the field may be identified in other ways, most frequently by their surface abundances. Despite the wide variety in terminology (e.g., Ba dwarfs and giants, CEMP-*s* and CH dwarfs and giants, dC stars) (see also **Supplemental Appendix Section 3**), high-resolution spectroscopy has revealed that these stars all share the property that their surfaces are enriched in carbon and *s*-process elements. They can all be understood within a single framework of mass gain from a companion AGB star that underwent thermal pulses and produced these elements by internal nucleosynthesis.

It is primarily the initial mass and metallicity, as well as the current evolution state, of the mass gainer that determine its appearance. For example, a metal-poor star more easily becomes C-enriched than a metal-rich star. If the surface C/O ratio increases above unity, strong bands of CH (and possibly other C-rich molecules) appear in the spectrum. In more metal-rich stars, strong Ba absorption lines often become the most notable feature. Thus, the distinction between Ba-, CH-, and CEMP-*s*-selected BSSs is primarily caused by decreasing metallicity. If the gainer mass is sufficiently high, it can evolve off the MS and become a similarly enriched field RGB or RC star.

approximate mass range of 2–3 $M_\odot$ over a wide range of accreted masses. To reproduce the stars with the highest *s*-process overabundances, accreted masses up to 0.5 $M_\odot$ are needed, which requires ≈20% of the mass lost by the AGB star to be accreted by the companion star (Stancliffe 2021). The results of Cseh et al. (2022) suggest that the highest mass-transfer efficiencies occur in Ba-star systems with current orbital periods $\lesssim$1,000 days. In a similar vein, Abate et al. (2015a) find that metal-poor AGB donors with masses in the range of ≈1–2 $M_\odot$ transferring up to 0.3 $M_\odot$ to their companions are needed to reproduce the abundances of a set of well-characterized C-rich BSSs and YSSs. As a note of caution, the constraints derived in the above and similar studies are necessarily dependent on the AGB nucleosynthesis models used, which contain many uncertainties. Lastly, Pal et al. (2024) detected sodium and *s*-process (barium, yttrium, and lanthanum) excesses in the M67 BSS WOCS 9005 and concluded that this BSS formed as the result of wind mass transfer from an AGB star. Specifically, they find that 0.15 $M_\odot$ was transferred onto a 0.45-$M_\odot$ MS progenitor companion.

Supplemental Material >

Ba excesses have also been found in other open cluster BSSs (Milliman et al. 2015, Nine et al. 2024). Nine et al. (2024) analyzed BSS Ba abundances in the open clusters NGC 7789, NGC 6819, and M67, spanning 2.4 Gyr. Some 15 of 35 BSSs in these three clusters show enhanced Ba, with a possible increased enrichment with younger formation age. The Ba-enriched BSSs lie in the same limited regions of the HRD irrespective of cluster age (**Figure 5**). Nine et al. note the increasing separation of the Ba-enriched BSSs from the cluster MSTOs with increasing cluster age and suggest that less massive AGB donors tend to undergo more efficient mass transfer. Some of the Ba-enriched BSSs are in long-period binaries, but surprisingly most are not detected as being velocity variable. Nine et al. hypothesize that these are binaries in periods longer than the detection limit of 10^4 days, indicative of AGB-driven wind mass-transfer origins.

RGB mass-transfer products are less easily identified by abundance measurements. Mass-gainer surface-abundance diagnostics—primarily CNO ratios—are more difficult to measure and may have limited lifetimes. Interestingly, Ferraro et al. (2006) observed evidence for C and O depletion in a subset of 6 BSSs in the globular cluster 47 Tuc. However, as two of these BSSs are W UMa binaries presumably comprised of two MS stars, Ferraro et al. suggest that these CO

W UMa binary: short-period eclipsing binary consisting of two low-mass main sequence stars in physical contact

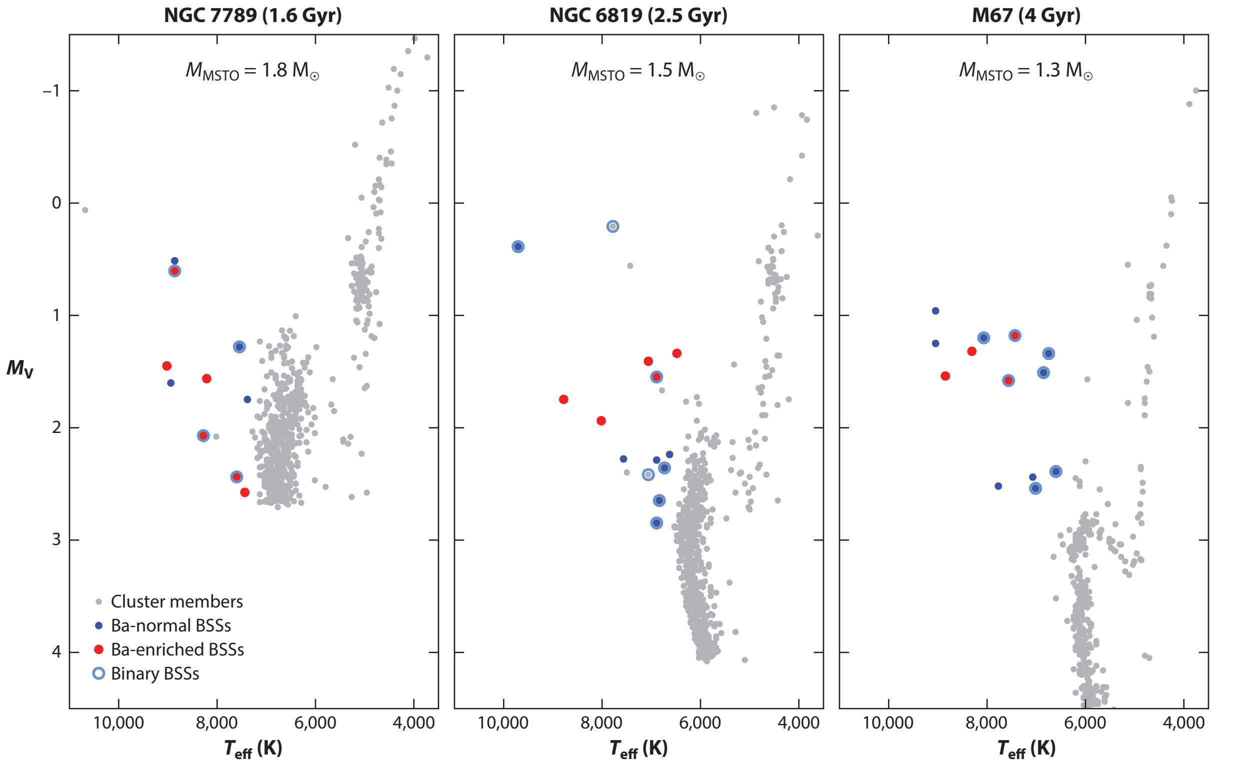

Figure 5

Hertzsprung–Russell diagrams of three open clusters showing Ba-enriched BSSs in (*a*) NGC 7789, (*b*) NGC 6819, and (*c*) M67. Blue points show the BSSs observed for Ba abundances; points highlighted in red show the BSSs that are enhanced in Ba compared to the MS. BSSs in known binary systems are indicated with a blue circle. Figure adapted from Nine et al. (2024) (CC BY 4.0). Abbreviation: BSSs, blue straggler stars.

depletions reveal CNO processed material accreted from deep within MS donor cores (see also Lovisi et al. 2013).

Lithium is a potentially interesting abundance diagnostic for distinguishing between BSS formation mechanisms, although the predictions are not secure. Carney et al. (2005) note that five of their six binary halo BSSs are depleted in lithium (although one lies in the lithium gap), which they argue is consistent with mass transfer. As yet, attempts to detect lithium in the M67 BSSs have yielded only upper limits (e.g., Shetrone & Sandquist 2000). Shetrone & Sandquist conclude that these limits are consistent with a formation process without mixing.

3.6. Rotation

Fundamentally a result of angular momentum conservation, rapidly rotating mass gainers are a prediction of every origin story of first-stage binary evolution. In fact, rapid rotation is a frequent, and indeed sometimes identifying, property of binary evolution products. The initial M67 BSSs identified by Johnson & Sandage (1955) have long been known to be rapidly rotating. Preston & Sneden (2000) and Carney et al. (2005) measured projected rotation velocities of halo BSSs, finding significantly higher $v \sin i$ values for binaries than for non-velocity-variable stars. The late-type MS companions to long-period sdB stars have $v \sin i$ values between 8 and 80 k s^{-1} (Vos et al. 2018). These rotations are suggestive of significant prior mass transfer from the progenitor RGB star. If this is in fact the case, these rapidly rotating companions would be classified as BSSs.

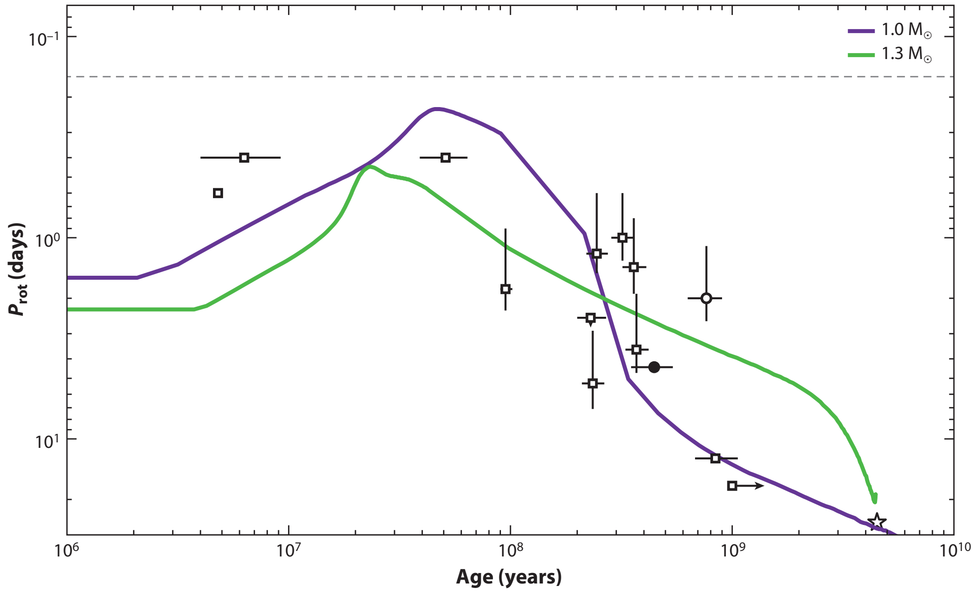

Figure 6

Age–rotation diagram of field BSSs and NGC 188 BSSs (*open squares*) and an M67 blue lurker (*filled circle*), all with hot WD companions from which the ages are derived. Uncertainties of 1σ are shown with black bars. Arrows indicate where the WD age or rotation period is a limit. The curves trace spin evolution models for 1.0-$M_\odot$ and 1.3-$M_\odot$ single stars (shown in *purple* and *green*, respectively) from Amard et al. (2019). The dashed black line shows the critical rotation period for a 1-$M_\odot$, 1-$R_\odot$ star from Ekström et al. (2008). The age and rotation period of the Sun are shown with an open star symbol for reference. Figure adapted from Nine et al. (2023) (CC BY 4.0). Abbreviations: BSSs, blue straggler stars; WD, white dwarf.

Leiner et al. (2018) examined the rotation periods of young NGC 188 BSSs as a function of transformation age (measured from cooling ages of WD companions), combining them with three even younger BSSs and two older BSSs in the Galactic field (**Figure 6**). The youngest BSSs with transformation ages of less than 10 Myr have rotation periods within a factor of a few of breakup rotation, which is consistent with expectations from all BSS formation mechanisms. The older BSSs (transformation ages as long as 1 Gyr) show spin-down rates typical for solar-like stars. This finding suggests that the magnetic fields and winds of late-type BSSs are typical of similar MS stars. Furthermore, this opens the opportunity for gyrochronological dating of BSSs.

In globular clusters, the history of rotation measurements has been more varied. Early observations revealed very rapidly rotating BSSs (Shara et al. 1997, De Marco et al. 2005). Observations of BSSs in other clusters found low $v \sin i$ measurements to predominate and suggested an inverse correlation between density and BSS rotation velocity. In contrast, Simunovic & Puzia (2014) conclude that rapidly rotating BSSs preferentially form in the cores of globular clusters.

Based on a study of projected rotation velocities of 320 BSSs in 8 globular clusters, Ferraro et al. (2023) find significant environmental effects on BSS rotation. They find the fraction of fast rotating BSSs ($v \sin i > 40$ k s^{-1}) to decrease dramatically with increasing density (**Figure 7**) and, specifically, by one order of magnitude for the observed ranges in central density, global (photometric) binary fraction, collision frequency, and state of dynamical evolution (precore collapse). Ferraro et al. suggest that the lack of rapid rotators in high-density environments is due to a dynamical reduction of the binary fraction limiting mass-transfer formation, combined with an unknown efficient spin-down mechanism associated with collisionally formed BSSs.

Breakup rotation: rotation rate at which the centrifugal acceleration balances gravity at the stellar equator, usually computed assuming the Roche model

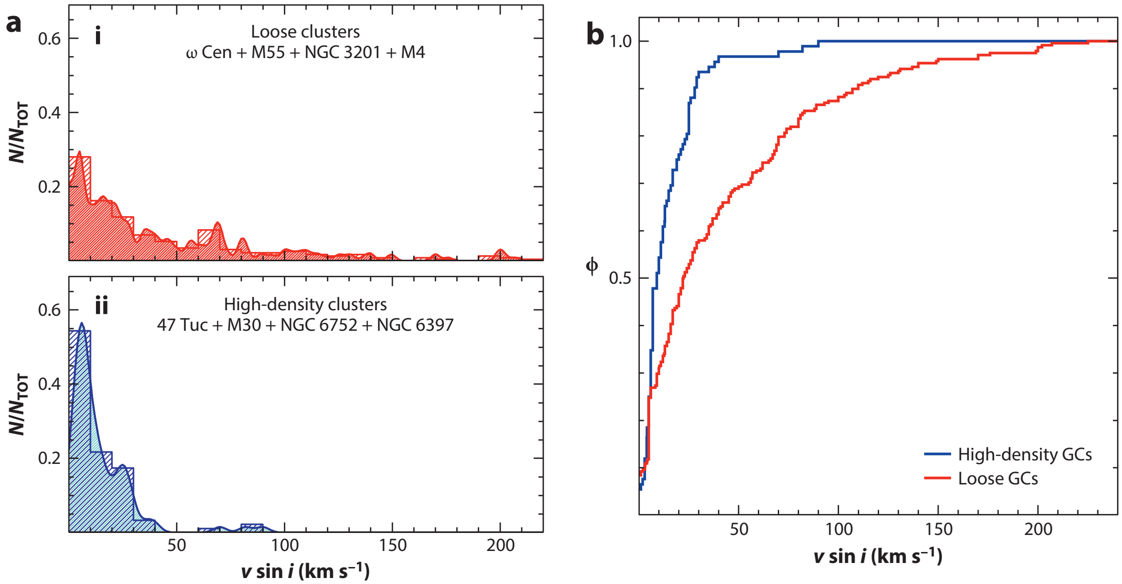

Figure 7

(*a*) Comparison of BSS projected rotational velocity histograms and kernel-density distributions for (*i*) low- and (*ii*) high-density GCs. In both cases, the fraction is relative to the total number of BSSs observed in each sample. (*b*) The normalized cumulative distributions. They differ at more than the 5σ significance level. Figure adapted from Ferraro et al. (2023) (CC BY 4.0). Abbreviation: BSS, blue straggler star; GCs, globular clusters.

Finally, Kaluzny et al. (2013) and Billi et al. (2023) find within globular clusters that BSS rotation tends to decrease for lower surface temperature and luminosity.

3.7. Environmental Effects in Globular Clusters

Globular clusters represent laboratories of environmental effects on first-stage binary evolution. In comparison to even rich open clusters, an evident difference of Galactic globular clusters is their much greater number of stars ($\approx 10^5$ compared to $\approx 10^3$). In addition to greater numbers of stars, globular clusters often have much higher central densities. Combined with their greater ages, globular clusters can be much more dynamically evolved and, as such, are important laboratories for studying alternative evolutionary pathways that are dynamical in origin, including stellar collisions (e.g., Pooley et al. 2003). In addition to these environmental differences, the larger ages of globular clusters also imply lower-mass progenitor stars for their binary evolution products, with typical MSTO masses of $\approx 0.7\,M_\odot$, and very low metallicities. Finally, globular clusters are often more distant, which can make their study more challenging technically.

Supplemental Material >

There have been extensive searches for connections between classical BSS frequencies and globular cluster properties, often yielding ambiguous results (**Supplemental Appendix Section 2.2.1**) (Knigge 2015). However, in globular cluster cores Knigge et al. (2009) found a positive correlation of the number of BSSs with core mass, from which they conclude that most BSSs, even in cluster cores, form from binary systems rather than from single-star collisions. In addition, Sollima et al. (2008) and Milone et al. (2012) found correlations between BSS specific frequencies and photometric-binary fractions in the cores of globular clusters of varying densities.

In the core of the low-density cluster NGC 3201, Giesers et al. (2019) directly measure spectroscopically a BSS binary frequency of 58 ± 8%, which is somewhat lower than that found in open clusters and the Galactic halo but statistically consistent. As also found in open clusters, this frequency is substantially higher than both their observed binary frequency of 20 ± 0.2% for all NGC 3201 core stars and their derived total binary frequency of 6.7 ± 0.7% for the cluster. Müller-Horn et al. (2025) also find the observed BSS binary frequency to be three times higher than the MS binary frequency in the core of the dense globular cluster 47 Tuc but with much smaller frequencies. Every indication is that the formation of BSSs in globular clusters is closely linked to the binary populations, which themselves vary in frequency between clusters (**Supplemental Appendix Section 2.2.3.1**).

Supplemental Material >

Interestingly, Giesers et al. (2019) were able to constrain the orbital parameters of 3 of 11 BSS binaries in NGC 3201 to periods of less than one day and small minimum companion masses (0.11 $M_\odot$ to 0.24 $M_\odot$). One of these binaries is also a contact system. (However, Müller-Horn et al. (2025) find a dearth of such short-period binaries in 47 Tuc.)

Perhaps related, photometric time-series studies have also revealed rich populations of very-short-period W UMa binaries among BSSs in globular clusters. Rucinski (2000) found that the W UMa frequency among BSSs in globular clusters is three times greater than that among field dwarfs. Such a high frequency is not found in open clusters (Rucinski 1998).

Lastly, globular clusters show a marked lack of short-period sdB binaries, in contrast to field sdBs (Moni Bidin et al. 2009, 2011; Latour et al. 2018). When combined with the high frequency of W UMa binaries and the presence of very-short-period BSSs binaries, there is an intriguing suggestion of interrelated phenomena occurring among close binaries in the globular cluster environment.

One of the most striking findings in globular clusters has been a double sequence of BSSs in the CMD of M30 (Ferraro et al. 2009) (**Figure 8**). The numbers of BSSs in both sequences are comparable. Notably, the BSSs in the red sequence are more centrally concentrated and have one of the highest specific frequencies in any globular cluster. Ferraro et al. hypothesize that the blue sequence represents BSSs created collisionally, whereas the red sequence formed (and may still be forming) from mass transfer, based on theoretical models of both processes (Sills et al. 2009, Xin et al. 2015) (see **Supplemental Appendix Section 2.2.2**). They further hypothesize that both populations were formed in a recent core collapse of the cluster. Subsequent papers have suggested that a few other globular clusters have similar BSS double sequences, with varying degrees of clarity. Curiously, W UMa variables—presumably mass-transfer products—have been found within the blue sequences of both M30 and NGC 362.

In summary, globular clusters show multiple differences in their binary evolution products between each other and with open clusters and the Galactic field. The relative influences of differing progenitor binary populations (e.g., due to dynamical evolution) and differing properties of the stars themselves (lower masses and metallicities) remain an important open question.

3.8. Red Stragglers and Sub-Subgiants

RSSs and SSGs are recent additions to the landscape of binary evolution, and the roles that they play are not yet clear. As with the other binary evolution products, they are identified by CMD locations distinct from single-star MS and giant branch stars. Furthermore, given their very high short-period-binary frequency, there seems to be little doubt that their CMD positions are a consequence of their binarity. What is less certain is the physical nature of that connection and, in particular here, whether mass gain or loss is involved.

A complete literature census of SSGs and RSSs in open and globular clusters was conducted by Geller et al. (2017a), finding 43 SSGs and 7 RSSs with secure cluster memberships. Although many more examples have been noted subsequently, in both globular clusters (Göttgens et al. 2019)

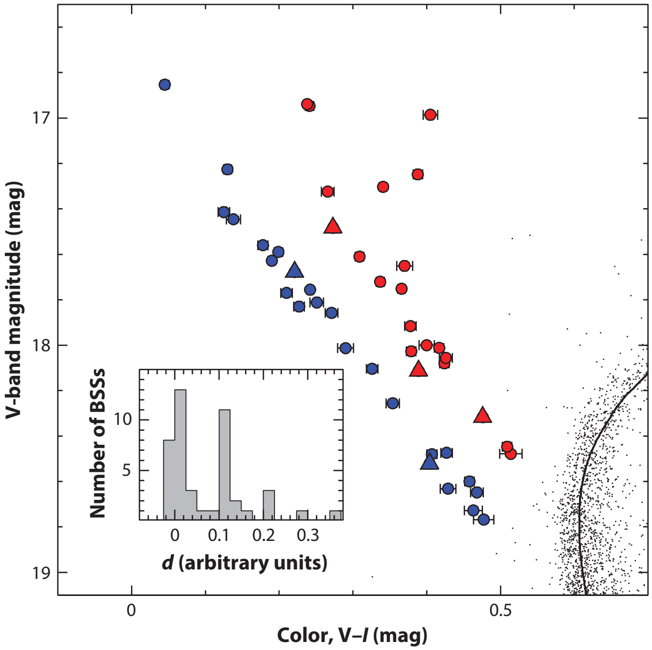

Figure 8

Color–magnitude diagram of the BSS region of *M*30. The red and blue colors distinguish the red and the blue BSS sequences, respectively. On the basis of their light curves, the three brightest variables have been classified as W UMa contact binaries (*triangles*). (*Inset*) Distribution of the distances, *d*, of the BSSs from the straight line that best fits the blue BSS sequence. Figure adapted from Ferraro et al. 2009 with permission from Springer Nature. Abbreviation: BSS, blue straggler star.

and especially the Galactic field (Leiner et al. 2022), the summative findings of Geller et al. have changed little.

Photometric variability is a defining property of both SSGs and RSSs. More than half have detected X-ray emission, and Hα emission is also frequent. Ca H and K emission has also been seen. Periodic photometric variability with periods $\leq$15 days is characteristic of SSGs, presumably due to spot modulation indicative of rotation with equivalent periods. Combined, these observations indicate substantial stellar activity linked to the rapid rotation of these late-type stars with convective envelopes.

Comprehensive radial-velocity surveys in four open clusters show short-period radial-velocity variability in 8 of 11 SSGs and in 1 RSS. Six orbit solutions have periods of less than 20 days. The shorter orbital periods match the photometric periods, indicating synchronous rotation. Giesers et al. (2019) also found short-period orbit solutions for four SSGs or RSSs identified in the globular cluster NGC 3201. These too show evidence of stellar activity, and in one case possibly accretion. All of the determined orbits to date are circular, except for the two longest period systems (17–18 days). Two double-lined systems—one SSG and one RSS—have MS companions.

Geller et al. (2017a) note a diversity of companions among the cluster SSGs, including two black hole/neutron star candidates and at least one millisecond pulsar. Three are in W UMa binaries. Albrow et al. (2001) find one of the SSGs in the globular cluster 47 Tuc to be a 1.1-day-period eclipsing binary with an optically identified WD and a UV excess, which they cite as a long-period cataclysmic variable candidate. These cases suggest that mass transfer or dynamical formation channels are possible.

However, Leiner et al. (2017) argue that most SSGs have reduced luminosities due to their enhanced magnetic activity and associated inefficient convection. Building on this hypothesis, Leiner et al. (2022) searched for SSG candidates among 1,723 active giant binaries (RS CVn–type variables) in the Galactic field. They identified 408 SSG candidates falling below or to the red of a 14-Gyr isochrone. (By the taxonomy here, the latter are RSSs.) Almost all (95%) of the candidates have rotation periods of 2–20 days, with a higher yield at shorter rotation periods. They conclude that these stars are a typical outcome of giant evolution in close binaries due to the associated rapid rotation.

Building upon an unbiased sample of 7,286 field RGB stars from APOGEE (Apache Point Observatory Galactic Evolution Experiment) DR17 with measurable rotation, Dixon et al. (2025) identify a subset of 38 SSGs similarly located relative to a 14-Gyr isochrone (and so these are also RSSs by our taxonomy). They find these stars to show high magnetic activity indices and to be rapidly rotating, which they attribute to tidal synchronization with as yet undetected binary companions.

3.9. Summary of Observed Properties

1. First-stage binary evolution products are frequent throughout upper HRDs for stars with masses less than 2 $M_\odot$. BSSs form a continuous sequence from classical BSSs to stars embedded in the MS (BLs), all having similar rotation, binary, and/or surface abundance properties. YSSs are found both to the blue of and embedded within giant branches. SSGs are distributed below subgiant branches, and RSSs are distributed to the red of RGBs. Finally, sdBs are found far to the blue of MSTOs.
2. Highly precise and accurate dynamical mass measurements for these products are few, but available mass determinations show classical BSSs and YSSs to have masses greater than the MSTO of coeval populations.
3. Unbiased surveys of the products show high binary frequencies that are far above the binary frequencies among host-population MS stars.
4. Most classical BSSs—even those very recently formed (e.g., having transformation ages less than 100 Myr)—do not lie on the ZAMS, which suggests some evolved cores at formation.
5. Long orbital periods (hundreds to many thousands of days) are frequent. There is a marked drop in frequency at periods below ≈100 days. These products show a wide range of orbital eccentricities from circular to $e \approx 0.8$ at long periods, with some dependence on period. Less frequent but important cases at very short periods (days or less) are also found.
6. The products are rapidly rotating compared to host-population MS stars. Very rapid rotation approaching breakup has been observed in the youngest BSSs, with subsequent spin-down at rates that are typical for solar-like stars.
7. Numerous lines of observational evidence show mass transfer to be the most frequent formation channel for first-stage binary evolution products. Orbital periods of thousands of days, surface abundance variations of *s*-process elements, and/or C/O WD companions indicate AGB mass transfer. Orbital periods of hundreds of days, in combination with He WD companions, and He-burning sdB stars indicate RGB mass transfer. W UMa stars are evidence for MS mass transfer.
8. Distributions of BSS CMD locations and surface abundance analyses provide evidence for a range of mass-transfer efficiencies.
9. Non-velocity-variable products and MS companions may result from merger or collision formation channels. Observed W UMa stars are likely progenitors of merger formation, whereas dependencies of globular cluster products on core density may indicate collisional formation in binary resonant encounters.

10. Globular clusters show differences in their product populations, which is suggestive of environmental influences. These include BSS double sequences in CMDs, variations in BSS and MS binary frequencies, inverse dependence of BSS rotation on density, high frequencies of W UMa stars, and a paucity of short-period sdB binaries relative to the field.
11. SSGs have short binary periods (on the order of days). Rapid rotation (in some cases tidally synchronized) and chromospheric indicators (X-ray emission, Hα, and Ca H and K emission) are consistent with substantial stellar activity.

4. THEORETICAL UNDERSTANDING OF FIRST-STAGE BINARY EVOLUTION PATHWAYS

4.1. Overview of Currently Hypothesized Formation Mechanisms

Our understanding of how the variety of objects and their properties presented in Section 3 can be produced by the evolution of binary stars starts with an appreciation of the initial conditions. About one-third of all solar-type stars have one binary companion, whereas a smaller fraction are in triples (~8%) or higher-order multiples (~3%) (Duquennoy & Mayor 1991, Raghavan et al. 2010). In the past decade, it has become increasingly clear that these frequencies strongly increase with primary mass and that the distributions of orbital periods and mass ratios are also mass dependent (Duchêne & Kraus 2013, Moe & Di Stefano 2017).

We consider three first-stage binary evolution channels: mass transfer, mergers, and collisions. The formation rate of each channel depends, in part, on progenitor binary frequency as a function of period. **Figure 9** shows an analytical approximation to the initial binary period distribution by Moe & Di Stefano (2017), using a 1.5-$M_\odot$ primary star as a representative example for the systems discussed in this review. The figure highlights the fractions of such stars that are expected

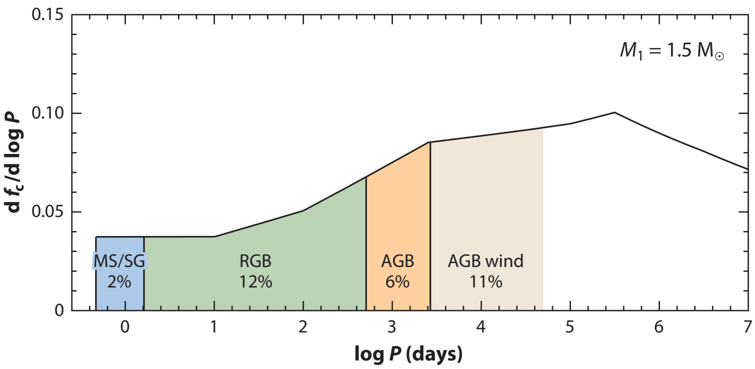

Figure 9

Initial distribution of binary orbital periods for a 1.5-$M_\odot$ star, showing the likelihood that such a star interacts with a companion during a certain phase of its evolution. The ordinate shows the frequency of companions (with mass ratio >0.1) per $\log_{10}P$ interval for a 1.5-$M_\odot$ primary, according to Moe & Di Stefano (2017). The vertical lines are the periods at which a donor fills its Roche lobe at the zero-age main sequence ($P \approx 0.4$ days), the base of the RGB ($P \approx 1.5$ days), the tip of the RGB ($P \approx 500$ days) and the tip of the AGB ($P \approx 3{,}000$ days), respectively, according to a MIST 1.5-$M_\odot$ solar-metallicity stellar model. Also shown is the longer-period range in which significant mass transfer via an AGB wind is expected (the upper limit of which is taken to be where Bondi–Hoyle–Lyttleton (BHL) accretion theory predicts at least 1% of the mass lost by the primary to be accreted by the companion). The percentages indicate the fraction of all 1.5-$M_\odot$ stars that will interact with a companion during the indicated evolution phase. Abbreviations: AGB, asymptotic giant branch; MIST, MESA (Modules for Experiments in Stellar Astrophysics) Isochrones and Stellar Tracks; MS, main sequence; RGB, red giant branch; SG, subgiant.

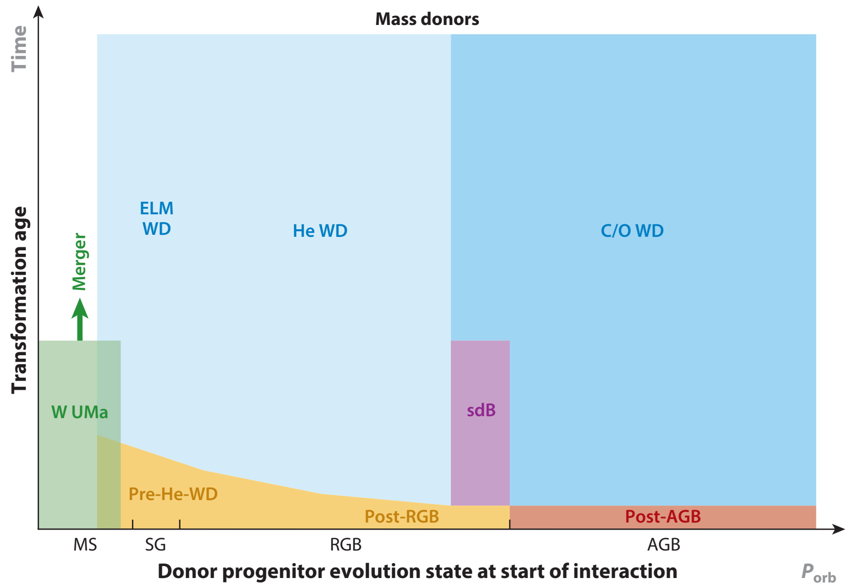

Figure 10

Various types of mass donors, schematically shown on a diagram of transformation age (on an arbitrary, nonlinear scale) versus the evolution state of the donor star at the beginning of the interaction. The abscissa also corresponds to the initial orbital period (also on an arbitrary, nonlinear scale) and can be correlated with **Figure 9**. Abbreviations: AGB, asymptotic giant branch; ELM, extremely low mass; MS, main sequence; RGB, red giant branch; sdB, subdwarf B; SG, subgiant; WD, white dwarf.

to interact with a lower-mass companion during different phases of their stellar evolution, using Roche-lobe-filling separations as the physical benchmarks.

Figures 10 and **11** schematically depict how observed binary evolution products can be produced by mass loss and mass gain, respectively, through mass transfer and mergers. In both figures, products are arranged by the stellar evolution state of the donor progenitor star when the interaction started (which in turn is related to the initial orbital period) and the time elapsed since interaction, i.e., the transformation age.

In the mass-transfer channel, the vast majority of interactions comprise the transfer of mass from an evolved donor onto a largely unevolved MS gainer. Upon completion, the remnant donor core remains as a companion to the mass gainer, with mass and composition set by the evolutionary state of the donor star during the mass transfer. Typically these remnant cores evolve into WDs through a variety of observational guises (**Figure 10**). In the merger channel, a more massive and evolved donor star typically merges with an unevolved MS star. We call the single star resulting from the merger the mass gainer. Such a mass gainer starts its new lease on life in a more evolved state corresponding to that of the donor star before the merger (as indicated in **Figure 11**). In either channel, after completion the mass gainers continue their evolution along single-star stellar evolutionary sequences.

About 20% of all 1.5-$M_\odot$ stars undergo mass transfer by RLOF, with a further ~10% expected to experience significant wind mass transfer (**Figure 9**). The vast majority will interact while the donor is a giant, on either the RGB or AGB. The reasons are two-fold: a low-mass star undergoes most of its radius expansion as an RGB or AGB star, and the initial period distribution is weighted toward the wider orbits in which only a giant star can fill its Roche lobe or initiate substantial

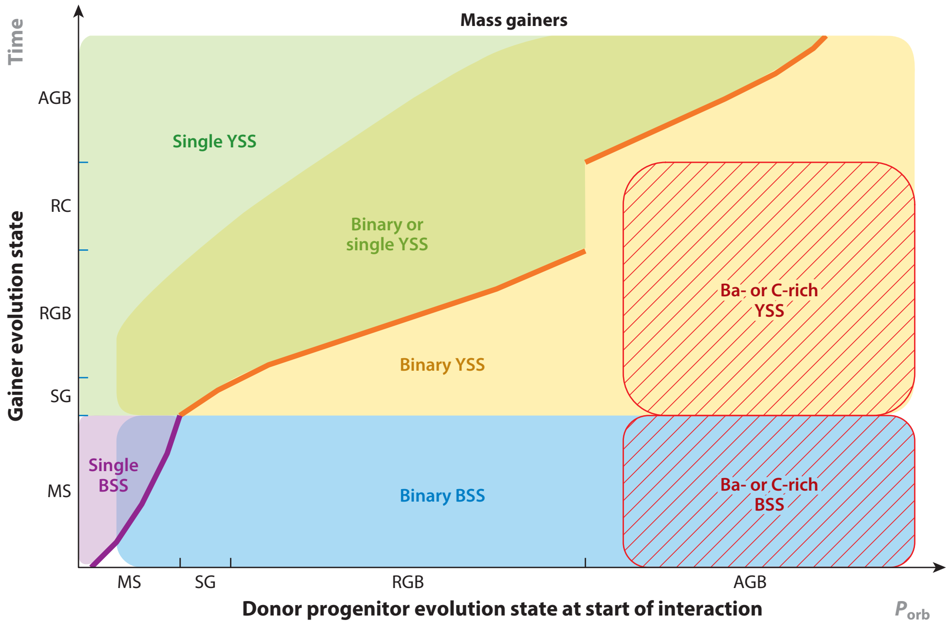

Figure 11

Similar to **Figure 10** but showing the various types of mass gainers discussed in this review. The ordinate, though still a time axis, now also shows the evolution state of the mass gainer (and cannot be directly compared to the ordinate of **Figure 10**). We distinguish mass gainers resulting from mass transfer, indicated as "binary," and mass gainers resulting from mergers (or collisions), indicated as "single"—although both can have additional companions if they formed in higher-order multiples. The purple and orange lines indicate approximately at which evolution state merger products emerge from the interaction. Abbreviations: AGB, asymptotic giant branch; BSS, blue straggler star; MS, main sequence; RC, red clump; RGB, red giant branch; SG, subgiant; WD, white dwarf; YSS, yellow straggler star.

wind mass transfer. We should thus expect mass transfer from a giant progenitor to be a frequent evolution channel, as observations show.

The stability of mass transfer and the mass-transfer efficiency play important roles in determining the amount by which the gainer can grow in mass as well as the evolution of the orbit. Both are a matter of uncertainty and debate, as is discussed in Section 4.2. Stable mass transfer is a self-regulated process, typically occurring on the donor's global thermal or nuclear timescale, which may allow the companion star to gain substantial mass and the orbit to either modestly shrink or (more commonly) significantly expand.

Unstable mass transfer, by contrast, is a runaway process occurring on the donor's dynamical timescale. This is thought to produce a CE surrounding both stars, in which energy and angular momentum are transferred from the orbit to the envelope, leading to a rapid spiral-in. If the envelope can be successfully ejected, a short-period binary emerges composed of the donor's original core and the companion. Currently, the latter is thought to have little or no mass gain. If the envelope is not ejected, the binary system must merge into a single object, which is effectively a mass gainer when compared to the original two stars. CE evolution has been the topic of numerous reviews [notably Ivanova et al. (2013, 2020), to which we refer the reader for details]; in what follows, we only address certain aspects relevant to this review.

There are multiple pathways to merger products (Section 4.3). Mergers of two MS stars can be initiated by expansion of the primary radius or by orbital angular momentum losses (such as magnetic braking), resulting in the formation and eventual merger of a W UMa-type contact binary. The merger product is a BSS. However, if the more evolved star had already developed a dense, hydrogen-free core, the merged object also has a post-MS core-envelope structure, so it will appear as a YSS. Merger products are depicted as single BSS and single YSS in **Figure 11**, even if they may have distant companions.

Hierarchical triple: triple system in which the tertiary orbits at sufficiently large distance from the inner binary such that the system can be decomposed into two near-Keplerian orbits

Systems of three or more stars are dynamically stable only if they are hierarchical. Much the same interaction processes as discussed above for isolated binaries are also expected to occur in the inner binaries of higher-order multiple systems. In addition, the orbits in a hierarchical triple or multiple can evolve on a secular timescale as they exchange angular momentum through Lidov–Kozai (LK) cycles. This allows mass transfer and mergers to take place in conditions not encountered in binaries (Section 4.2.4). Furthermore, the orbits in a hierarchical multiple can become dynamically unstable as a result of secular orbital changes, leading to collisions. Thus triples and higher-order multiples, while not as common as binaries, can provide unique evolutionary pathways and add further complexity in defining the origins of specific binary evolution products.

In the middle of the last century, stellar collisions were considered to occur between single stars and only in very dense stellar environments. However, with the advent of N-body simulations of clusters and the seminal work by Heggie (1975), it became clear that in the presence of substantial binary populations, collisions within resonant dynamical encounters of binaries (or higher-order multiples) with other cluster stars can substantially increase the collision frequency (McMillan 1986, Leonard 1989). Fundamentally, this is because the encounter cross-section increases from stellar radii to binary separations. As with mergers, the result can be a product with higher mass than either of the two colliding stars, and in an evolutionary state set by the more evolved of the colliding stars (Sills et al. 2005).

4.2. Mass Transfer

In the context of mass-transfer formation, the existence of binary evolution products with a wide range of orbital periods suggests that a stable form of mass transfer, allowing for significant accretion by the secondary, must occur across the range of possible evolution states of the donor indicated in **Figures 10** and **11**. In particular this must hold for giant donors in order to explain BSSs and YSSs with periods between ∼100 and many thousands of days, as well as sdB binaries and post-RGB and post-AGB binaries with similar periods. The often very wide orbits of Ba-rich BSSs and YSSs, combined with the expectation that RLOF from AGB donors is in most cases unstable, led Boffin & Jorissen (1988) to suggest the now commonly adopted idea of AGB wind mass transfer for their origin. At the same time, populations of short-period WD-MS binaries and sdB-MS binaries ($P \sim 1$ day or less) indicate that mass transfer from giants also often results in a CE and spiral-in. Finally, the common occurrence of eccentric orbits among these binary evolution products goes against expectations from tidal interaction theory and requires an explanation. In this section, we review our understanding of the main issues playing a role: the stability and efficiency of mass transfer, its role in spin-up of the secondary, and the evolution of the orbital period and eccentricity during mass transfer.

4.2.1. Stability of mass transfer. The canonical way of describing the stability of RLOF (Webbink 1985, Soberman et al. 1997) is based on a linear stability analysis of the response to mass loss of the stellar radius versus that of the Roche-lobe radius. In addition, the initial stellar response is assumed to be adiabatic because its dynamical timescale is usually much shorter than its thermal timescale. Many BPS models use the stability criterion of Hjellming & Webbink (1987), which treats giant stars as condensed polytropes to approximate their compact cores and extended convective envelopes. This approximation results in stellar expansion in response to mass loss unless the core mass makes up a relatively large fraction of the total mass. The Hjellming & Webbink criterion predicts unstable mass transfer from RGB or AGB donors under almost all circumstances, except for a very small corner of parameter space. Indeed, populations of short-period binaries are readily reproduced by such BPS models as resulting from CE evolution, but many of those same models predict a conspicuous dearth of post-mass-transfer orbits with 100 days $\lesssim P \lesssim$ 1,000 days (e.g., Nie et al. 2012; see also Leiner & Geller 2021).

Detailed binary evolution models have been known to result in stable mass transfer under a wider range of circumstances than expected from the Hjellming & Webbink criterion (e.g., Han et al. 2002, Podsiadlowski et al. 2002, Chen & Han 2008b). Stellar-wind mass loss prior to RLOF as well as mass and angular-momentum loss during mass transfer can both contribute to greater stability. More fundamentally, in the outer envelopes of evolved giants convection is inefficient at transporting energy, and these layers can thermally relax on a timescale faster than the dynamical timescale (Woods & Ivanova 2011). These effects allow an evolved giant to contract even at mass-loss rates that are much faster than the global thermal timescale. This has been confirmed by recent detailed calculations over a wide parameter space by Ge et al. (2015, 2020) and Temmink et al. (2023). The latter authors find that RLOF from extended giant (RGB and early AGB) donors can be self-regulated and avoid a runaway over a wide range of mass ratios, albeit at very high mass-loss rates (intermediate between the global thermal and dynamical timescales) and in some cases at the expense of the giant overflowing the potential surface associated with its outer Lagrangian point.

Although uncertainty remains about how mass transfer proceeds in this regime, these studies indicate that stable RLOF from extended RGB stars is possible for a sizable fraction of binaries, in particular for relatively wide orbits. This is indeed required to explain the observed orbital periods of hundreds to about 1,400 days for sdB binaries and BSSs with He-WD companions and also may play a role in explaining the lack of such systems with $P <$ 100 days (Section 3.4).

RLOF mass transfer from thermally pulsing AGB stars remains almost unexplored by detailed binary evolution modeling. The physical situation (and any attempts at modeling it numerically) is complicated by the rapid radius excursions during thermal pulses, the very tenuous outer layers of these stars, and their dynamical (Mira) pulsations. Pastetter & Ritter (1989) studied stable RLOF from AGB donors that had already lost more than half of their initial mass by a stellar wind, such that $M_{\mathrm{donor}}/M_{\mathrm{gainer}} < 0.7$ (see also the sidebar titled WOCS 4540: A Long-Period BSS in NGC 188). The populations of Ba- and C-rich BSSs and YSSs with periods between 100 and ~3,000 days suggest it is worthwhile to explore whether a stable form of AGB RLOF can operate over a wider range of mass ratios.

4.2.2. Efficiency of mass transfer and spin-up of the companion.

How efficiently mass can be transferred from one star to another is expected to depend strongly on the mode of interaction: wind mass transfer, stable RLOF, or unstable RLOF. We discuss each of these regimes in turn. Before doing so, however, it is helpful to distinguish various aspects of the mass-transfer process.

β_{capt}: fraction of mass lost by the donor star that is dynamically captured by the gravitational field of the companion

Of the mass lost by a donor star, a certain fraction (β_{capt}) may be dynamically captured within the gravitational sphere of influence of the companion. Only in short-period binaries (roughly $P \lesssim$ 10 days for low-mass MS companions; see Ulrich & Burger 1976) does this captured mass fall directly onto the companion. Owing to the angular momentum of the gas stream, in wider binaries it forms an accretion disc from which matter gradually spirals toward the companion surface. The matter that approaches the companion surface is not guaranteed to be accreted, as this depends on the star's response to accretion, which may expel part of the material.

An important factor for understanding mass-transfer efficiency is the angular momentum accreted together with the mass, which very efficiently spins up the companion (Section 4.2.2.4). As a result of this and potentially other factors, the fraction of mass lost by the donor that is actually

WOCS 4540: A LONG-PERIOD BSS IN NGC 188

WOCS 4540 is the longest-period BSS-WD pair in the 7-Gyr open cluster NGC 188, with a period of 3,030 days and an eccentric orbit ($e = 0.36$). The WD mass of $0.53 \pm 0.03\ M_{\odot}$ indicates it is a CO WD. It is also one of the most luminous BSSs in the cluster, requiring a mass gain of at least $0.3\ M_{\odot}$.

Sun & Mathieu (2023) used `MESA` (Modules for Experiments in Stellar Astrophysics) to reproduce the system in detail, requiring a combination of RGB wind mass transfer, AGB wind Roche-lobe overflow (WRLOF), and AGB RLOF during a thermal pulse. All three mechanisms contribute similar masses to create the 1.5-$M_{\odot}$ BSS. The integrated mass-transfer efficiency is 55%, rising to 75% during a thermal pulse yielding a very large but brief and stable accretion rate. The modeling did not include evolution of stellar spins or orbital eccentricity.

Given the physical assumptions of the simulation, the evolution was well determined. The model began with two ZAMS stars having masses of 1.2 and 1.14 $M_{\odot}$ in a circular orbit with a period of 1,330 days. Efficient wind mass transfer during the RGB phase is required; otherwise with standard wind mass loss parameters, the RGB wind so depletes the donor envelope that insufficient mass remains to create the BSS. AGB WRLOF is effective but not sufficient to produce a BSS of sufficient mass. This mass requirement sets the high RLOF mass-transfer efficiency during the largest thermal pulse. Strong stellar winds in the late RGB and AGB phases after mass reversal expand the orbit, which allows the RLOF efficiency to be as high as 75% and still be stable. This model demonstrates the possibility of forming long-period BSS binaries by stable mass transfer, if the required high mass-transfer efficiencies can be realized in nature.

accreted by the companion, β_{acc}, may be significantly smaller than β_{capt}. Hydrodynamical simulations of mass transfer can inform β_{capt}, but not β_{acc}, which depends on slower, and often uncertain, evolutionary processes. Furthermore, exactly how the mass that is lost leaves the system affects the orbital evolution (Section 4.2.3), which feeds back onto the mass-transfer rate and accretion rate.

As a result of these difficulties, the overall efficiency of mass transfer and how it varies over binary parameter space have remained outstanding challenges to binary evolution theory. Observational constraints (Section 3) provide important clues, but a full picture of how β_{acc} depends on binary parameters is still lacking. See also the sidebar titled BSSs from Stable Mass Transfer.

4.2.2.1. Wind mass transfer. Stellar wind mass transfer (see Boffin 2015 for a review) has often been approximated as BHL accretion (Hoyle & Lyttleton 1939, Bondi & Hoyle 1944). The BHL accretion efficiency strongly decreases with increasing ratio of wind velocity to orbital velocity. For the relatively slow winds of RGB and AGB stars, BHL accretion predicts modest β_{capt} values of at most 10% (when $v_{\mathrm{wind}} \sim v_{\mathrm{orb}}$). However, AGB winds are accelerated only at distances from the star where dust can form, which is typically a few stellar radii. When the wind acceleration occurs outside the donor Roche lobe, hydrodynamical simulations show that the stellar wind is funneled through the L1 point toward the companion star (Mohamed & Podsiadlowski 2007). In this regime of wind Roche-lobe overflow (WRLOF), simulations find β_{capt} values up to 30–50% (Abate et al. 2013; Saladino et al. 2019; Chen et al. 2017b, 2020). The accretion rates measured in these simulations are model- and resolution-dependent and often are strongly time variable; the quoted values are time-averaged.

The hydrodynamical simulations cited above typically find that the captured wind material carries significant angular momentum. Simulations that take cooling of the gas into account show that this gas accumulates in an accretion disk (Chen et al. 2017b, Saladino et al. 2018). The concomitant angular momentum transfer poses a possible limitation to the actual mass-transfer efficiency (Section 4.2.2.4).

β_{acc}: mass-transfer efficiency, i.e., fraction of mass lost by the donor star that is accreted by the companion ($\beta_{\mathrm{acc}} \leq \beta_{\mathrm{capt}}$)

BSSs FROM STABLE MASS TRANSFER

The formation of BSSs in M67 via stable mass transfer in primordial binaries was modeled by Tian et al. (2006). Their models predict only a few BSSs, far fewer than observed. The products are all binaries with very short orbital periods (<10 days). Only 2 out of 19 M67 BSSs, S1082 and S1284 with orbital periods of less than 4 days, fall in the period range of their products. Also seeking to model M67, Lu et al. (2010) follow both case A and case B of mass transfer, considering progenitor primary stars with masses between 1.3 $M_{\odot}$ and 1.6 $M_{\odot}$. They find that case B formation rates equal those of case A and merger origins combined. They find that mergers tend to produce more luminous BSSs. They also show that nonconservative mass transfer produces lower-mass BSSs, whose longer lifetimes yield more BSSs. With a range of mass-transfer efficiencies, they can create the CMD positions of all M67 BSSs (except F81) but again cannot produce their number.

More recently, Leiner & Geller (2021) applied the population synthesis code `COSMIC` to reproducing the BSS populations in 16 old open clusters, including M67. Varying a wide range of RLOF stability criteria based on mass ratio, they too were unable to reproduce the BSS numbers. They estimate that mergers and collisions cannot make up these differences in full and conclude that mass transfer from giant donors must be more stable than typically assumed. [Geller et al. (2013) came to the same conclusion for the NGC 188 BSS population.] Also, the `COSMIC` models produce more high-mass BSSs than observed but do not produce enough BSSs near the MSTO. Leiner & Geller again note that lower mass-transfer efficiency might increase the relative numbers of lower-luminosity BSSs. That no population syntheses yet reproduce the number of BSSs in old open clusters is likely an important insight into gaps in our current formation theories.

4.2.2.2. Stable Roche-lobe overflow. During stable RLOF on a nuclear or thermal timescale, the matter transferred via the inner Lagrangian point does not have sufficient energy to escape from the Roche lobe of the companion, such that one may expect $\beta_{\mathrm{capt}} \approx 1$. This suggests the possibility of fully conservative mass transfer, in which all mass lost by the donor is gained by the companion. However, for more rapid mass transfer (such as from a Roche-lobe filling giant), the donor radius can substantially exceed its Roche lobe and mass loss from the outer Lagrangian points may occur, implying $\beta_{\mathrm{capt}} < 1$. As for wind accretion, an important limiting factor to the actual mass-transfer efficiency (β_{acc}) may come from the effects of spin-up of the companion by angular momentum transfer.

A further limitation to mass gain can arise when the companion expands to fill its own Roche lobe as a result of mass transfer. This is another situation that can lead to mass loss from the outer Lagrangian points or, alternatively, to a merger via a CE phase, in particular when the initial mass ratio is far from unity (Section 4.3.1). Theoretical studies of the response to accretion have mostly focused on fairly massive (>2 $M_{\odot}$) MS stars with radiative envelopes (e.g., Kippenhahn & Meyer-Hofmeister 1977), finding that such stars swell up appreciably when accreting at rates faster than dictated by their thermal timescale. However, low-mass ($\lesssim$1.5 $M_{\odot}$) MS stars with convective envelopes are able to accommodate high accretion rates with only modest or no expansion (Fujimoto & Iben 1989, Zhao et al. 2024).

Although some binary evolution studies attempt to take these effects into account explicitly by following the evolution of both stars, it has been common practice in binary evolution calculations to treat the mass-transfer efficiency as a free parameter during RLOF. See the sidebar titled WOCS 5379: A BSS from Stable Red Giant Branch Mass Transfer.

4.2.2.3. Unstable mass transfer. During the CE phase that results from unstable mass transfer, negligible mass gain by the secondary is expected on theoretical grounds. Ivanova et al. (2013) point out that significant accretion during a CE phase is unlikely because of the much larger

WOCS 5379: A BSS FROM STABLE RED GIANT BRANCH MASS TRANSFER

WOCS 5379 in the 7-Gyr open cluster NGC 188 is a BSS in a 120-day period eccentric orbit, with a precisely measured mass of $0.42 \pm 0.02\ M_\odot$ for a He WD companion. Sun et al. (2021) used the stellar evolution code `MESA` (Paxton et al. 2011) in the approximation of a circular orbit to reproduce the system in some detail with RGB mass transfer, at an efficiency of 22% to yield the observed mass and orbital period. Their progenitor binary pairs a 1.19-$M_\odot$ star and a 1.01-$M_\odot$ star in a 12.7-day circular orbit; the current MSTO mass is 1.2 $M_\odot$. Importantly, mass transfer began with a very rapid phase on a thermal timescale, during which half of the transferred mass occurs. The rapid mass transfer was then slowed by the consequent orbit expansion, followed by a long period at a lower mass-transfer rate. Finally, although the transformation age is 300 Myr, after mass gain the interior structure of the BSS matches that of a 1.2-$M_\odot$ star with an evolution age of 2.1 Gyr. Importantly, neither orbital eccentricity nor accretor spin evolution was included.

Sun et al. (2021) were not able to simultaneously reproduce the measured WD mass and effective temperature (cooling age). The modeled WD mass was 0.33 $M_\odot$. They suggested resolution via a variable mass-transfer rate, permitting additional time for the donor core growth. The 0.42-$M_\odot$ mass is greater than that expected from the orbital period-WD mass relation for the products of stable RGB mass transfer (Rappaport et al. 1995) (**Figure 12**). In that context, reproducing the WD mass would require the primary to have filled its Roche lobe in a wider orbit than assumed in the model, with a period of about 100 days, which is similar to the current period. This would imply that mass transfer happened with much less orbital expansion than expected for stable RLOF.

entropy of the CE compared to the surface layers of an MS star. Indeed 3D hydrodynamical CE simulations typically find negligible accretion (Ricker & Taam 2008, 2012).

This conclusion appears to be at odds with the fact that a significant population of dC stars has been found in binaries with $P < 4$ days (Section 3.4.2). Miszalski et al. (2013) estimate that, in the case of the short-period dC star binary inside the Necklace planetary nebula (**Supplemental Appendix Section 3.2.3**), between 0.03 and 0.35 $M_\odot$ must have been accreted by the dC star to explain its high C abundance. They argue that this must be the result of mass gain before the CE phase, likely as a result of wind accretion from the former AGB companion. This is supported by the observation of jets, likely launched by accretion onto the companion, that are apparently older than the nebula itself. A similar scenario may explain other C-rich BSSs in short-period binaries.

Supplemental Material >

4.2.2.4. Spin-up by mass transfer. Packet (1981) showed analytically that accretion from a Keplerian disk can spin up an accreting star to breakup rotation after gaining only about 5–10% of its original mass. In an attempt to study the amount of mass that can be accreted by the progenitors of CEMP-*s* and Ba stars, Matrozis et al. (2017) showed with detailed modeling that accretion of about 0.05 $M_\odot$ from a Keplerian disk spins up these stars to breakup. This is <5% of their original mass; the difference with the analytical estimate comes from taking structural changes during the accretion into account. Applying these findings, Vos et al. (2018) argued that the MS companions of wide sdB binaries accreted less than 0.01 $M_\odot$ during RGB mass transfer in order to explain their observed rotation velocities.

If β_{acc} is truly limited by spin-up to very low values, this poses a serious problem for the formation of wide-orbit BSSs by mass transfer (whether by wind accretion, WRLOF, or RLOF, because all of these processes are expected to produce accretion disks). As discussed in Section 3 and in the sidebars titled WOCS 4540: A Long-Period BSS in NGC 188 and WOCS 5379: A BSS from Stable Red Giant Branch Mass Transfer, multiple observations indicate a wide range of mass-transfer efficiencies that extend well above 10%.

It therefore seems likely that companions are able to accrete much more efficiently than inferred from the spin-up limit. Only in short-period binaries ($P \sim$ days or less) is this problem less severe, because disk accretion can be avoided, and tidal interaction may be strong enough to counteract spin-up. One possible resolution is suggested by Popham & Narayan (1991), who have shown that when rotation is close to breakup, angular momentum can be transported outward from the star to the disk by shear stresses, provided there is sufficient viscous coupling between the star and the accretion disk. Another mechanism may be provided by magnetic torques that tie the stellar spin to the disk rotation, as suggested for accreting pre-MS stars (e.g., Armitage & Clarke 1996).

Spin-up limit: limit to the amount of mass gain imposed by accretion leading to breakup rotation

4.2.3. Orbital evolution and the *P–e* diagram.

The orbital period–eccentricity diagrams of BSSs, YSSs, and other binary evolution products (**Figure 4**) provide important clues to their evolutionary origins. This section reviews our current understanding of binary orbital evolution.

Changes in the orbital period, semimajor axis, and eccentricity of a binary are induced by a number of processes operating before, during, and after mass transfer. From angular momentum and energy conservation arguments, analytical expressions for such orbital changes can be derived under certain simplifying assumptions [e.g., see Soberman et al. (1997) for mass loss and mass transfer in circular orbits and Sepinsky et al. (2007, 2009) and Dosopoulou & Kalogera (2016) for an extension to eccentric orbits]. These relations can inform, for example, under which circumstances orbital expansion or shrinking is expected. In addition to mass loss and mass transfer, dissipative tidal interactions lead to both orbital circularization and spin synchronization.

4.2.3.1. Orbital period evolution.

A well-known result of orbital evolution theory is that fully conservative mass transfer shrinks the orbit if mass is transferred from a more massive to a less massive star, and vice versa. Conservative mass transfer typically inverts the mass ratio to such an extent that the final mass ratio is more extreme (further from unity) than the initial value, leading to a net expansion of the orbit (e.g., Han et al. 2000).

The situation is less clear for nonconservative mass transfer. The outcome in this case depends not only on the mass-transfer efficiency ($\beta_{\rm acc}$) but also on the amount of angular momentum carried by the mass that is lost from the binary system. A common assumption for nonconservative RLOF, known as isotropic reemission, is that mass leaves the system in the form of a fast isotropic wind or a symmetric pair of jets from the vicinity of the accreting star, as is expected (and often observed) for accretion by compact objects. This matter then takes away the specific orbital angular momentum of the companion, which widens the orbit to a similar extent as conservative mass transfer (Soberman et al. 1997). Although wind- or jet-like outflows from accreting MS stars have been observed, for example, in post-AGB binaries (Van Winckel 2019), the mechanism by which they are generated and the circumstances under which they can occur are poorly understood.

- **Mass loss and wind mass transfer**. A commonly used result of the analytical approach to orbital evolution is that fast, isotropic mass loss from one or both stars (also known as adiabatic mass loss) in the absence of mass transfer always expands the orbit (with the period following $P \propto M_{\rm tot}^{-2}$) and leaves the eccentricity unchanged. BPS models that apply this model yield period distributions of Ba- and C-rich BSSs and YSSs that overproduce long periods (>10,000 days) and underproduce periods <2,000 days (e.g., Izzard et al. 2010, Abate et al. 2015b).

  However, the underlying assumptions of the isotropic wind model break down for the slow winds of giants, in particular for AGB stars. The modified flow pattern and spiral-shaped accretion wake induced by the gravitational field of the companion exert a torque on the binary that enhances the orbital angular momentum loss compared to the isotropic case, as has been shown in hydrodynamical simulations (Jahanara et al. 2005; Saladino et al. 2018, 2019; Chen et al. 2018, 2020). The effect on the orbital period has been quantified by Saladino et al. (2019), showing that the orbit shrinks rather than expands when the wind velocity is less than the relative orbital velocity. More recent simulations including dust formation by Chen et al. (2020) indicate that this conclusion holds irrespective of whether mass transfer is in the BHL or the WRLOF regime. Abate et al. (2018) show that adopting the results of hydrodynamical simulations in BPS models helps to produce more CEMP-*s* stars in the period range of 100–2,500 days but not enough to reproduce the observed period distribution.

- **Stable Roche-lobe overflow.** A testable prediction of stable RLOF from low-mass RGB donors is that the post-mass-transfer orbital period should follow a tight relation with the final WD mass (Rappaport et al. 1995, Lin et al. 2011). This relation results from the core-mass radius relation of RGB stars with degenerate He cores, together with the expectation that the radius closely follows the Roche-lobe radius during stable mass transfer. These conditions make the relation somewhat dependent on metallicity (because metal-poor RGB stars have smaller radii for the same core mass) but independent of whether RLOF was conservative or nonconservative (see, e.g., Chen et al. 2013). The relation extends to the larger WD masses and longer periods applicable to AGB donors, although it is less tight in that regime because of thermal-pulse-related radius variations.

  The P–M_{WD} relation allows a test of whether certain systems indeed result from stable RLOF. Chen et al. (2013) found that models of stable RLOF starting near the tip of the RGB can reproduce the orbital period distribution of wide sdB binaries and that these systems should similarly follow a P–M_{sdB} relation, although sdB masses are not measured accurately enough to test this prediction. **Figure 12** shows how the P–M_{WD} relation compares with several other binary samples having measured orbital periods and WD masses. The two NGC 188 BSSs are discussed in the sidebars titled WOCS 4540: A Long-Period BSS in NGC 188 and WOCS 5379: A BSS from Stable Red Giant Branch Mass Transfer. Parsons et al. (2023) reported the discovery of three WD-subgiant binaries with periods between 40 and 460 days, in which the WD masses (0.24–0.40 $M_\odot$, each $\pm 0.01\,M_\odot$, determined from HST spectroscopy) are consistent with the relation; the somewhat discrepant object is the inner binary of a triple system. Garbutt et al. (2024) studied a sample of FGK stars with WD companions (found from UV excess flux) and periods in the range of 100–1,000 days for which *Gaia* DR3 astrometric orbits are available, allowing a determination of the WD mass. At the short-period end, EL CVn–type eclipsing binaries consist of a low-mass pre-He-WD and an A- or F-type MS companion (van Roestel et al. 2018); Chen et al. (2017a) show that these systems can be formed by stable RLOF starting from a low-mass MS or subgiant donor. In these four samples, the WD masses cluster around the values expected from the P–M_{WD} relation in the respective period range, but with a large scatter. Some (though not all) of the more discrepant objects have larger errors than the mean values indicated in **Figure 12**. By and large, the data support a stable RGB mass-transfer origin for most of these objects.

  In contrast, the products of AGB mass transfer do not appear to adhere to such a P–M_{WD} relation. Some of the WDs in the Garbutt et al. (2024) sample have masses $\gtrsim 0.6\,M_\odot$; their periods are too small to have resulted from stable RLOF (see also Yamaguchi et al. 2024). The Ba- and C-rich BSSs have a wide range of orbital periods, straddling the values of a few thousand days expected from stable RLOF (see **Figure 12**). Although the larger periods can be accounted for by wind mass transfer, periods less than about 2,000 days appear to require substantial angular momentum loss during RLOF.

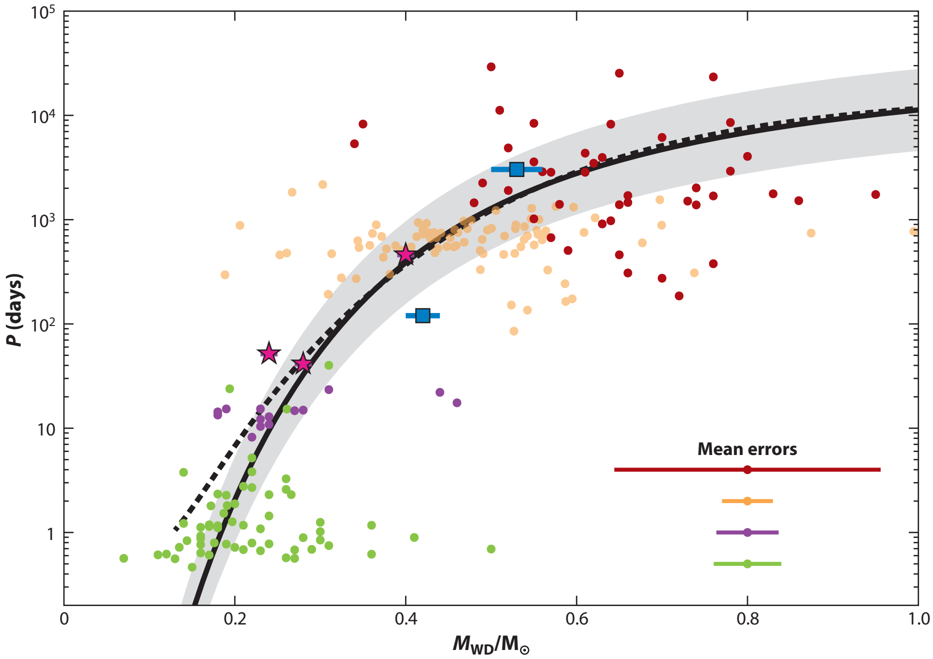

Figure 12

Periods and WD masses for two BSSs in NGC 188 (WOCS 4540 and WOCS 5379, *blue squares*), three WD-subgiant binaries from Parsons et al. (2023, *pink stars*), Ba-rich BSSs and yellow straggler stars from Escorza & De Rosa (2023, *red dots*), FGK+WD binaries with *Gaia* astrometric orbits from Garbutt et al. (2024, *yellow dots*), and EL CVn–type binaries from van Roestel et al. (2018, *green dots*) and El-Badry & Rix (2022, *purple dots*). For the latter four samples, the error bars on the WD masses often are substantial; mean errors are shown in the lower-right corner. The solid line is the P–M_{WD} relation from Lin et al. (2011), updated from Rappaport et al. (1995, *dotted line*). The gray band is a factor of 2.4 uncertainty as proposed by Rappaport et al. (1995). Abbreviations: BSSs, blue straggler stars; WD, white dwarf.

One possibility is that nonconservative mass transfer is accompanied by much stronger angular momentum loss than in the isotropic reemission model. For example, the nonaccreted mass may leave the binary via the outer Lagrangian points, perhaps after accumulating in the Roche lobe of the companion. Matter that leaves via L2 or L3 potentially forms a circumbinary torus or disk around the binary (e.g., Frankowski & Jorissen 2007), which may further influence the orbital evolution and in particular the eccentricity (Section 4.2.3.2).

- **Unstable mass transfer.** An alternative possibility for the formation of AGB mass-transfer products with periods of hundreds of days is an efficient form of CE ejection, in contrast to the spiral-in process needed to form short-period (days or less) post-CE binaries. Such a scenario was found by Abate et al. (2015a,b) to be able to produce C-rich BSSs with periods of a few hundred days, although in a much smaller proportion to longer-period systems than is observed. In their models, these stars acquired their elevated C and *s*-process abundances from AGB wind mass transfer prior to the CE phase. More recently, Yamaguchi et al. (2024) found efficient CE ejection from AGB donors to be a viable formation channel for wide-orbit (40–300 days) WD-MS binaries discovered from *Gaia* astrometry.

On the other hand, as noted in Section 4.2.2.3, short-period C-rich BSSs are also plausibly formed by a CE occurring after a phase of wind mass transfer from an AGB donor. This requires a very inefficient CE ejection process, as shown by Abate et al. (2015a) in the case of the CEMP-*s* dwarf HE 0024-2523 with $P \approx 3$ days. It is not clear what would make CE ejection efficient in some cases and very inefficient in others. Recent hydrodynamical simulations of CE evolution with AGB donor stars indicate post-CE orbital periods that are intermediate between those of the short-period and long-period C-rich BSSs (Sand et al. 2020, González-Bolívar et al. 2022).

4.2.3.2. Eccentricity evolution. Tidal interactions are expected to be efficient at circularizing orbits of low-mass binaries before RLOF occurs (Verbunt & Phinney 1995). Thus, systems that have gone through a phase of mass transfer have been expected to be circular, including in BPS studies. Different implementations of the viscous tidal damping timescale can give quantitatively different results (Preece et al. 2022) but do not change this overall picture.

The observation of frequent noncircular orbits, and in particular the discovery in the mid-1990s of binary post-AGB stars with eccentric orbits and periods on the order of a year, has prompted investigation of mechanisms to increase the eccentricity during or after mass transfer. These fall into three main categories: mass-loss and mass-transfer variations along the orbit, interaction between the binary and a circumbinary disk, and other proposals.

- **Phase-dependent mass loss and mass transfer.** Following a suggestion by Van Winckel et al. (1995), Soker (2000) showed that bursts of mass loss from an AGB star at periastron passage in a binary orbit can counteract tidal circularization under some circumstances, as can mass transfer at periastron if it occurs from a less massive donor to a more massive companion. Bonačić Marinović et al. (2008) expanded on this idea by proposing a model for tidally enhanced mass loss from AGB stars, varying smoothly with distance and peaking near periastron. They find that, when combined with a somewhat reduced tidal dissipation rate, this model can keep the orbit eccentric up to the point when the AGB star fills its Roche lobe at periastron and can account for the eccentricities of Ba stars with periods greater than ~1,000 days. Kashi & Soker (2018) proposed that, rather than by tidal effects, mass loss from an AGB star may be enhanced at periastron passage by accretion-induced jets from the companion as it grazes the envelope of the AGB star and, thus, also counteract tidal circularization.

  Enhanced mass loss is much less likely to explain the eccentric orbits of BSSs originating from RGB rather than AGB donors, because very strong mass-loss enhancement suppresses RLOF (Siess et al. 2014). Indeed, in their detailed modeling of sdB binaries, Vos et al. (2015) found it to prevent the formation of the sdB star altogether. They also considered RLOF at periastron and found this to be capable of increasing an initially small eccentricity (0.001) to $e \lesssim 0.15$. However, their widest orbits are all circularized, in contrast to observations.
- **Circumbinary disks.** The discovery of circumbinary disks around binary post-AGB stars led Waelkens et al. (1996) to suggest that their eccentric orbits may be the result of gravitational interaction with such a disk, as was proposed for pre-MS binaries (Artymowicz et al. 1991). An analytical model for the resonant interaction of a binary with a circumbinary disk was applied by Dermine et al. (2013) to post-AGB binaries and by Vos et al. (2015) to sdB binaries. They found that such a model can account for the range of observed eccentricities in both classes of systems, although the result is sensitive to highly uncertain disk parameters; these authors assumed a constant disk mass and lifetime. By contrast, Oomen et al. (2020) found only a modest increase in the eccentricities of post-AGB binaries (from 0.01 to $e \lesssim 0.2$) when taking into account additional constraints from the evolution of the post-AGB star as well as the counteracting effect of tides. This is not enough to reproduce observations.

Although there is no evidence of current circumbinary disks among binary BSSs and YSSs with similar periods and eccentricities, such disks may have existed during or shortly after the mass-transfer process and since dispersed. Disk–binary interaction may therefore have a wider applicability.

- **Other mechanisms.** The above mechanisms require a small seed eccentricity ($e \sim 10^{-3}$) to increase the eccentricity; they cannot make a circular binary eccentric. A possible mechanism without this limitation arises when a WD emerges from an AGB progenitor with a small initial (kick) velocity. Izzard et al. (2010) find that the eccentricities of Ba stars can be reproduced in BPS modeling with an assumed WD kick velocity of several kilometers per second. The physical origin of such a kick is unclear, and the WD must be accelerated on a timescale shorter than the orbital period. Circumstantial evidence for nonzero WD velocities at birth has been reported from WD radial distributions in globular clusters (Davis et al. 2008). Further possibilities to explain the existence of eccentric orbits among BSS binaries arise in triples and higher-order multiple systems (Section 4.2.4).

4.2.4. The effect of multiplicity on mass transfer. Tertiary companions are especially prevalent among the closest MS binaries: Tokovinin et al. (2006) found 96% of solar-type binaries with $P < 3$ days to have a tertiary companion. The possibility that triple systems may play a role in the formation of BSSs and YSSs is suggested by the discovery of three triple Ba-rich products (shown in **Figure 4**) (see Jorissen et al. 1998, Escorza et al. 2019b, Escorza & De Rosa 2023).

The presence of a third star can change the evolution of a binary system in several ways (see Toonen et al. 2016 and Naoz 2016 for reviews). Foremost among these effects is the possibility of LK cycles: secular oscillations in the eccentricity and inclination of the inner orbit. These cycles in turn lead to enhanced tidal interactions, mass transfer, and mergers and collisions in the inner binary (see Section 4.3).

Toonen et al. (2020) explored these and other effects in triples (with stellar masses between 1 and 7.5 $M_\odot$) using population synthesis. (Because primary masses were distributed according to the stellar initial mass function, their results are dominated by low-mass stars even though they include stars more massive than 2 $M_\odot$.) Toonen et al. find that mass transfer occurs in 66–77% of triples, which is about twice as frequently as in a population of binaries with the same masses. In the vast majority of cases, mass transfer starts in the inner binary. Furthermore, in about 40% of mass-transfer cases, the orbit is still eccentric when mass transfer starts, as LK cycles counteract the tides that would otherwise circularize the orbit. These effects are most prominent among the closest inner binaries, with the incidence of mass transfer among MS and subgiant donors increasing about fourfold compared to a binary population, and 90% of such binaries still being eccentric at the start of RLOF.

A rare but interesting possibility is mass transfer from the tertiary to the inner binary, which requires the tertiary to be the most massive star in the system and the outer orbit to be sufficiently close. Toonen et al. (2020) find that RLOF from the tertiary occurs in ∼1% of all triples, predominantly when it is in its AGB phase of evolution; they did not consider wind mass transfer, which would increase this fraction. De Vries et al. (2014) and Portegies Zwart & Leigh (2019) modeled the initial phases of such tertiary mass transfer using a combination of stellar evolution, stellar dynamics, and hydrodynamics. De Vries et al. find the accretion stream intersects the inner orbit, preventing the formation of an accretion disk, with most of the stream being ejected from the inner binary and causing the inner orbit to shrink. For a different parameter space, Portegies Zwart & Leigh (2019) find a variety of possible outcomes: orbital shrinking and eventual merger of the inner binary, or orbital expansion of the inner binary. In some cases the latter outcome causes the triple to become dynamically unstable, and in other cases it results in efficient accretion (via a circumbinary disk) onto both stars in the inner binary, which evolve toward equal masses. The latter scenario can potentially account for observed double-BSS or BSS-MS binaries, in at least one of which there are indications of a WD tertiary (**Supplemental Appendix Section 3.2.3**). Inner binary mergers induced by tertiary mass transfer were studied by Gao et al. (2023) in the context of Ba-rich BSSs.

Supplemental Material >

4.3. Mergers and Collisions

BSSs can be created through the merger or collision of two MS stars. In the literature, the terms mergers and collisions are often used interchangeably, but they are fundamentally different physical processes with different timescales. Mergers are physical encounters of two tightly bound stars leading to their contact and coalescence. Collisions are physical encounters of two stars unbound (or loosely bound) to each other. For given stellar masses, collisions are much more energetic encounters and occur on shorter (dynamical) timescales than mergers.

4.3.1. Mergers. Following on the suggestion of McCrea (1964), early papers exploring this formation path focused on mergers of contact binaries, resulting from case A mass transfer (e.g., Webbink 1976). Nelson & Eggleton (2001) explored conservative case A mass transfer across a wide parameter space, finding evolution into contact for most initial orbital periods and mass ratios. Building on this work, Chen & Han (2008a) modeled case A mergers of solar-like stars (0.89–2 $M_{\odot}$), considering only conservative mass transfer and avoiding processes on dynamical timescales (e.g., CE) and angular momentum losses. They find that their merger products may reproduce the more luminous BSSs in M67 (but they do not consider the observed binary status of those BSSs). Recently, Henneco et al. (2024) also explored the binary parameter space for evolution into contact, instead assuming highly nonconservative mass transfer by taking the spin-up limit into account. Their results are qualitatively similar to those of Nelson & Eggleton (2001) where they overlap. Contact is again found to be a frequent outcome of case A mass transfer and is expected to eventually lead to a merger and BSS formation. They also modeled wider binaries in which the donor is a giant and the mass-transfer rate can be so high that the companion swells up to fill its Roche lobe; this can be another route to mergers but creates evolved products with already formed He cores, i.e., YSSs (see also **Figure 11**).

Andronov et al. (2006) conducted population analyses to evaluate wind braking as an angular momentum loss mechanism leading to the merger of two low-mass ($M < 1.2$ $M_{\odot}$) MS stars in a close ($P < 5$ days) binary. They note significant differences in their wind braking timescales including saturation from findings of earlier works, especially for lower-mass binaries. Their models predict BSS numbers comparable to one-third of the numbers actually observed in open clusters, which they suggest to be comparable with the number of non-velocity-variable BSSs. Similar results are found for halo BSSs. Intriguingly, they hypothesize that mergers may be the primary source of BSSs in globular clusters. They estimate uncertainties in their predictions of a factor of two, convolving all of the assumptions in the initial binary populations.

Recently, substantial attention has been focused on BSS formation through mergers driven by tertiary stars (of both primordial and dynamical origin) through the combination of the LK mechanism and tidal friction. Perets & Fabrycky (2009) and Naoz & Fabrycky (2014) explore this mechanism in some detail, including evolution of triples in a collisional cluster environment. They find this channel to be plausible and testable, specifically by the presence of high binary frequencies among BSSs and, in particular, by long-period MS companions having noncoplanar spin-orbit orientations. The latter is not predicted by mass-transfer mechanisms but may not distinguish between merger mechanisms. Perets & Fabrycky (2009) especially argue that the e–log P diagram of the M67 and NGC 188 BSSs are naturally produced by this process, specifically the gap in the period distribution and the wide range of orbital eccentricities (**Figure 4**). Perets & Fabrycky (2009) note that signatures of this mechanism acting in the halo may be few, because the product BSSs would likely have evolved (and perhaps had a later mass-transfer event) in the lifetime of the halo. Naoz & Fabrycky (2014) make a similar point for clusters and require an ongoing creation of triples through collisional perturbations of primordial coplanar systems.

Recently, Shariat et al. (2025) have comprehensively modeled the role of triples in close binary evolution, including the formation of BSSs. Their models incorporate both three-body dynamics and stellar evolution, as well as parameterized mass transfer through the `COSMIC` BPS code. They find that most BSSs formed as mergers have companions at periods substantially greater than 3,000 days and, thus, predict that the majority of observed single BSSs formed as mergers in triples and harbor long-period companions (the original tertiaries). They note that the BSS binaries formed in this way naturally have a broad orbital eccentricity distribution, albeit more extended than that found in open clusters.

Finally, Antonini et al. (2016) explore merger formation of BSSs in dense cores of globular clusters as a result of dynamically formed triples. They find this channel to contribute less than 10% to BSS formation in globular clusters.

In these and other studies modeling the merger products, simplified assumptions are made about the structure and subsequent evolution of the merger product. Detailed calculations of the actual mergers in primordial binaries are challenging, because the timescales of the merger process are comparable with the stellar evolution timescales of the binary companions. Thus, detailed simulations must combine (magneto)hydrodynamics with stellar evolution. To date, such calculations have been done for binaries with stars more massive than considered here (Schneider et al. 2019).

4.3.2. Collisions. The classic paper by Hills & Day (1976) considered collisions between single stars for BSS formation in the densest cores of globular clusters. However, secure observational evidence for such single-star collisions has been difficult to find (e.g., Knigge et al. 2009). Collisions within encounters of multiple (especially binary) systems are likely more frequent and are now included in analyses of the BSS collisional formation channel.

Leigh & Sills (2011; see also Leigh et al. 2022) analyze the multiplicity interaction rates of collisional formation paths of BSS, paying particular attention to open clusters. **Figure 13** shows the rich variety of pathways leading to collisions in resonant encounters given a population of binaries and triples. Leigh and colleagues conclude that triple encounters can play a significant role in BSS formation, especially those BSSs in multiple systems, and specifically that most of the binary BSSs in the open cluster NGC 188 may form from dynamical encounters.

Monte Carlo techniques have been used extensively to model BSS formation in globular clusters, including the collision channel. One such body of work uses the `MOCCA` code, which is specifically designed to follow the evolution of objects formed in complicated dynamical interactions—both single-star collisions and resonant interactions—within the context of global cluster evolution (Hypki & Giersz 2013; see also Hypki & Giersz 2017b, and references therein). Hypki & Giersz (2017a) find that globular cluster initial conditions significantly affect the relative frequencies of BSSs formation pathways. Perhaps most importantly, they find that the relative frequency of mass-transfer formation (including mergers) and collision formation in binary encounters is very sensitive to the progenitor semimajor axis distribution. Prevalence of wide binaries favors high collision rates. In addition, higher initial central densities promote collisional formation channels, again primarily in binary encounters. In contrast, Hypki & Giersz (2017a) find that high densities have very little impact on the frequency of mass transfer or merger formation channels. A second body of Monte Carlo studies presented by Chatterjee et al. (2013) shows very similar results. Unfortunately, very little is known about the current semimajor axis distributions in globular clusters.

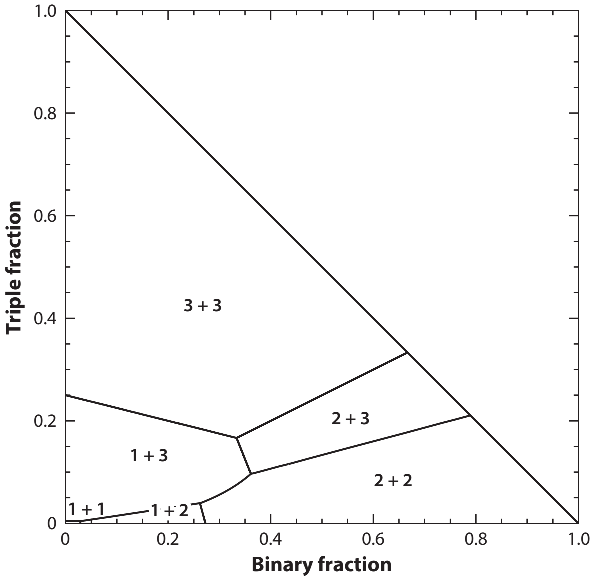

Figure 13

The parameter space in the binary fraction–triple fraction plane showing which of various encounter types dominate. Leigh & Sills (2011) assume that the binary semimajor axis is 90 R_* and the tertiary semimajor axis is five times greater. Figure adapted from Leigh & Sills (2011) with permission from OUP.

Perets & Kratter (2012) studied the consequences of hierarchical triples becoming dynamically unstable as a result of orbital changes in the inner binary due to mass loss and mass transfer, which Toonen et al. (2020) find to occur in 2–5% of triple systems. The subsequent dynamical evolution causes stellar collisions and exchanges and can form eccentric binaries with evolved stars (WDs) in close orbits. Collisions and exchanges in triples predominantly involve AGB stars.

Because collisions happen on dynamical timescales, they permit (magneto)hydrodynamical simulations. The products of these simulations have then been inserted into stellar evolution codes. Here, we briefly summarize the results of a substantial literature for BSSs specifically.

In addition to the stellar masses, the nature of the collision product depends sensitively on the impact parameter and the degree of mixing. Lombardi et al. (1995, 1996) and Sills et al. (1997) showed that the core of the more evolved star ends up in the core of the collision product, and the material of the less evolved star ends up on the outside of the collision product. Approximately, the materials in a collision product retain memory of their precollision entropy. Immediately after the collision, the product is not in thermal equilibrium. In the case of a collision of two MS stars, the initial postcollision evolution goes from a bloated state to a new MS position based on the core composition. This evolution occurs without further mixing (Glebbeek & Pols 2008), so the initial CMD position of the relaxed product, the BSS, is largely set by the core of the initially more evolved star (and thus need not be on the ZAMS).

Collision products are typically very rapidly rotating, depending on the impact parameter. Essentially, orbital angular momentum is converted to spin of the product. Given substantial conservation of angular momentum, after contraction the products may approach breakup speeds. Models of their spin-down depend on the prescriptions of subsequent angular momentum loss. These have been often tied to the presence of magnetic fields (unknown observationally, but see Ryu et al. 2025). Of particular interest is whether collision products are initially surrounded by disks that might permit magnetic disk-locked braking.

4.4. Models of Sub-Subgiants and Red Stragglers

Early speculations on the origins of SSGs focused on dynamical interactions in clusters (e.g., Albrow et al. 2001, Mathieu et al. 2003). Leiner et al. (2017) more deeply considered three pathways to SSGs in the open clusters M67 and NGC 6791: mass transfer, envelope stripping due to dynamical encounters, and reduced luminosities due to enhanced magnetic activity and inefficient convection.

Mass-transfer models were able to reproduce the CMD positions of these SSGs, but only for progenitor binaries of very short periods (≈1 day), greatly reducing the production frequency and yielding only short-period SSGs. Envelope stripping (modeled in one dimension as rapid mass loss) leads to nonthermal-equilibrium evolutionary paths through the SSG region, and subsequent equilibrium evolution through the RSS region. The production rate is set by the encounter frequency and is small in open clusters. Finally, rapid rotation and consequent enhanced stellar activity have been suggested to reduce the efficiency of envelope convection, and thereby lower luminosities. Stellar evolution models with arbitrarily reduced convection efficiency also take MSTO stars through the SSG region of the HRD (but can be somewhat challenged in the CMD) and into the RSS region. The probability of this pathway is much higher because it only requires rapid rotation, likely from tidal synchronization within a progenitor close-binary population.

Subsequently, Geller et al. (2017b) used Monte Carlo models of globular clusters to explore in more detail the formation frequencies along each of these pathways. Their analyses again argue for formation in isolated, tidally synchronized binaries, though all pathways appear to be viable (especially in higher-mass globular clusters). They also find subsequent population of the RSS domains, sometimes directly and sometimes as subsequent evolution from SSGs.

5. CLOSING THOUGHTS

The scope of this review is the first stage in the evolution of close, low-mass ($M < 2\,M_{\odot}$) MS binary stars. This is a major stellar evolution pathway; binary evolution products are a significant population relative to evolved single stars. Observations have greatly expanded our knowledge of these stars since prior comprehensive reviews, with a concomitant expansion in theoretical understanding. Summaries of what has been learned, and still needs to be learned, follow this section. Here, we close with high-level thoughts integrating current observations (Section 3) and theory (Section 4).

Classical BSSs and YSSs are now directly measured to have masses roughly consistent with masses determined from single-star evolutionary tracks. Based on such tracks applied in cluster HRDs, BSS masses are distributed from masses consistent with gains of entire MSTO stellar envelopes to having masses below the MSTO. Some BSSs and YSSs are also found with measured masses of twice or more their cluster MSTO masses.

An array of observations—including orbital periods, WD companions, and surface abundances—align with mass-transfer theory and have frequencies that point to mass transfer being the primary formation channel of binary evolution products. Furthermore, the mass-transfer paradigm is able to explain the many types of binary evolution products in a unifying way. Even so, many of the product masses are challenging to mass transfer in that theory has long predicted that even small mass gains will spin the gainers to above breakup rotation. In fact, observations show very recently formed BSSs to be rapidly rotating, albeit not at breakup. The companions of wide sdB binaries are also spinning fast. Whether there are products that reached breakup velocities during formation is unknown, as are their ensuing states. In any case, with respect to spin angular momentum the observed mass gains suggest that additional physics is needed.

Recent observations are also challenging for mass-transfer theory with respect to orbital evolution. In contrast to long-held expectations that mass-transfer products would have tidally circularized orbits, long-period BSSs and YSSs have a wide range of orbital eccentricities from circular to $e \approx 0.8$. In some cases WD companions confirm mass-transfer origins for these eccentric products. Initial theoretical work suggests that mass transfer may maintain or increase existing eccentricities. Dynamical effects from higher-order multiple systems or circumbinary disks may play a role as well. Much work regarding mass transfer in eccentric binaries remains to be done toward developing a holistic explanation of product orbits.

Long-standing theory has argued that RLOF mass transfer is unstable for most binaries with giant primaries. Yet products having the long-period orbits expected from RLOF mass transfer are frequent. For the widest orbits, resolution has been found through AGB wind mass transfer. In addition, more recent theory finds that RGB and AGB RLOF mass transfer is more stable than indicated by traditional binary stability criteria. Even so, observed orbital periods of hundreds of days are not easily produced by either stable RLOF (on a thermal or slower timescale, typically expanding the orbit) or unstable mass transfer (on a dynamical timescale, leading to a CE and strong orbit shrinkage). This may suggest the existence of an alternative mode of RGB or AGB mass transfer that acts on an intermediate timescale, which is a notion that finds some support from recent detailed modeling of RLOF stability domains.

Merger and collision formation channels also have long been considered. Merger products have been presumed to be the outcomes of W UMa contact binaries, and blue sequences in globular cluster CMDs have been suggested to have collision origins. However, specific merger and collision products are difficult to securely identify. Products that are apparently single, have large masses, or have MS companion stars are often cited as candidates. Simulations show that entropy sorting makes surface abundance signatures uncertain at best, at least for collision products. Asteroseismic studies of interiors may be a promising path forward (Rui & Fuller 2021), but currently there is no substantial sample of definite merger or collision products for detailed study and comparison with theory.

The presence of magnetic fields in products is unknown. However, rapid rotation is an expected outcome of all formation channels, and many of the binary products discussed here have convective envelopes, so dynamos may be expected. There is early observational evidence for BSS spin-down on timescales similar to those of single solar-like stars. Perhaps this is an indication of the formation and continued presence of magnetic fields in late-type binary evolution products. If so, they also may play an even earlier role in angular momentum redistribution, especially in the presence of disks or mass loss.

Observations in globular clusters show that the frequencies of binary evolution products are mediated by host binary frequencies. Numerical simulations show that binary frequencies are themselves mediated by stellar densities. Recent observations suggest that the properties of products also depend on stellar densities. These new connections remain to be understood.

Binary evolution products may also extend beyond the mass gain and loss paradigm. Tidal synchronization in close binaries yields rapid rotation of the components, which for observed single stars demonstrably affects stellar evolution including magnetic field strengths, effective temperatures, radii, and surface abundances. As evidenced by SSGs and perhaps RSSs, such products also populate domains of the HRD different from that of single stars.

The field of binary-star evolution is currently as fundamental, active, and exciting as it has ever been. Ultimately the goal is that comprehensive binary-evolution models should have detailed consistency with observations, including both the first phase of evolution reviewed here and subsequent evolution pathways.

SUMMARY POINTS

1. The products of binary evolution comprise a significant fraction of all evolved stars in older stellar populations. Their frequencies are broadly consistent with the fraction of low-mass stars expected to interact with a companion prior to the end of their giant evolution.
2. Classical blue straggler stars (BSSs) and yellow straggler stars have masses greater than the main sequence (MS) turnoffs of coeval populations. Multiple lines of observational evidence indicate mass transfer to be the predominant formation channel, occurring with a wide range of mass-transfer efficiencies.
3. All binary evolution products have high binary frequencies significantly above those of their host-population MS stars. Long orbital periods (hundreds to thousands of days) are frequent. Less frequent but important cases at very short periods (days or less) are also found. Long-period binaries show a dispersion of orbital eccentricities from circular to $e \approx 0.8$.
4. Many binary evolution products are rapidly rotating compared to host-population MS stars. Rapid rotation is a natural consequence of having gained mass and is an expected outcome of all mass transfer, merger, or collision formation. Excess angular momentum needs to be removed during formation, by processes that are yet to be fully elucidated, to permit observed mass gains.
5. Observed properties—orbital periods, white dwarf (WD) companions, surface abundances—point specifically to frequent origins in red giant branch (RGB) and asymptotic giant branch (AGB) mass transfer. Multiple lines of evidence (both observational and theoretical) suggest that such mass transfer is more stable than indicated by traditional binary stability criteria. The observed distributions of periods and eccentricities are not fully explained by current mass-transfer theory.
6. The products of merger and collision formation channels are difficult to securely identify but may be suggested by very-high-mass products, MS companions, or the absence of companions.
7. Most classical BSSs—even those very recently formed—do not lie near the zero-age MS. Their color–magnitude diagram (CMD) distributions bear information on the cores of progenitor secondary stars in mass-transfer formation or of progenitor primary stars in merger or collision formation, as well as on relative formation rates.
8. Sub-subgiants (SSGs) have short binary periods (on the order of days). Rapid rotation and chromospheric activity of SSGs are consistent with tidal synchronization. The CMD locations of both SSGs and red straggler stars (RSSs) can be explained with magnetically limited convection, as well as by stars that have been dynamically stripped.

FUTURE DIRECTIONS

1. Structures within comprehensive and unbiased orbital period distributions are likely to demarcate products of differing formation pathways [e.g., common envelope (CE) and stable Roche-lobe overflow]. The same may be true of the distributions of other orbital

parameters such as eccentricity and secondary mass, especially if all are integrated into a multidimensional analysis. Upcoming *Gaia* data releases will certainly be a foundation for such studies.

2. There remains a great need for comprehensive and detailed spectroscopic case studies of products. Especially, numerous detailed spectroscopic analyses of WD companions yielding masses and ages are crucial for constraining models of binary mass transfer.
3. The evolution of spin angular momentum during binary interactions has only nascent constraints from observations. More comprehensive stellar rotation distributions for binary products are vital, especially when combined with orbital parameters and young ages.
4. Understanding angular momentum transfer and evolution remains crucial for progress. Theoretical predictions of spin angular momentum gains during mass transfer, and indeed beyond breakup for the observed mass gains, must be resolved. Very likely such resolution will require deeper understanding of the role of accretion disks and (perhaps coupled) magnetic fields.
5. The stability of mass transfer from RGB and AGB donor stars remains ill-understood. Closely related, the nature and consequences of mass and angular momentum losses from the entire system during mass transfer are uncertain. These issues, as well as how mass transfer proceeds in eccentric orbits, will be integral for understanding observed distributions of orbital periods and eccentricities of mass-transfer products.
6. Certainly binary evolution is influenced by the presence of additional companions. The role of interactions within multiple systems is only beginning to be considered in a comprehensive way.
7. Detailed models for mergers of low-mass MS binaries are especially lacking and likely will require three-dimensional magnetohydrodynamic simulations. The amount of mass lost during the merger, the redistribution of angular momentum, and the postmerger structure and internal composition are still open questions.
8. There is no current theoretical framework for red stragglers. A theory for RSSs, perhaps integrated with theory for SSGs, will advance both the understanding and the classification of these stars.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

We gratefully acknowledge the generous assistance of many colleagues. We thank Evan Linck for invaluable and skilled technical support in creation of the manuscript. We appreciate the careful reading and insights of Alison Sills, Gijs Nelemans, and colleagues in the R.D.M. research group: Evan Linck, Erin Motherway, Andrew Nine, Jack Worley (University of Wisconsin); Nathan Leigh, Amanda Quitral Pierart, Magdalena Vilaxa Campos (University of Concepción); and Don Dixon (Vanderbilt University). We thank Emanuele Dalessandro, Ana Escorza, Aaron Geller, Alain Jorissen, and Hans van Winckel for valuable discussions.

We also gratefully acknowledge data and analyses in advance of publication from Elliott Horch, David Latham, Evan Linck, and Amanda Quitral Pierart and Nathan Leigh.

The seeds of this review were planted in a Biergarten in Garching during The Impact of Binaries on Stellar Evolution (ImBaSE) conference in 2017 and further nurtured during the follow-up ImBaSE 2022 conference in Garching. We are grateful to the organizers of both inspiring meetings.

R.D.M. was supported by the Wisconsin Alumni Research Fund and NSF AST-1714506.

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